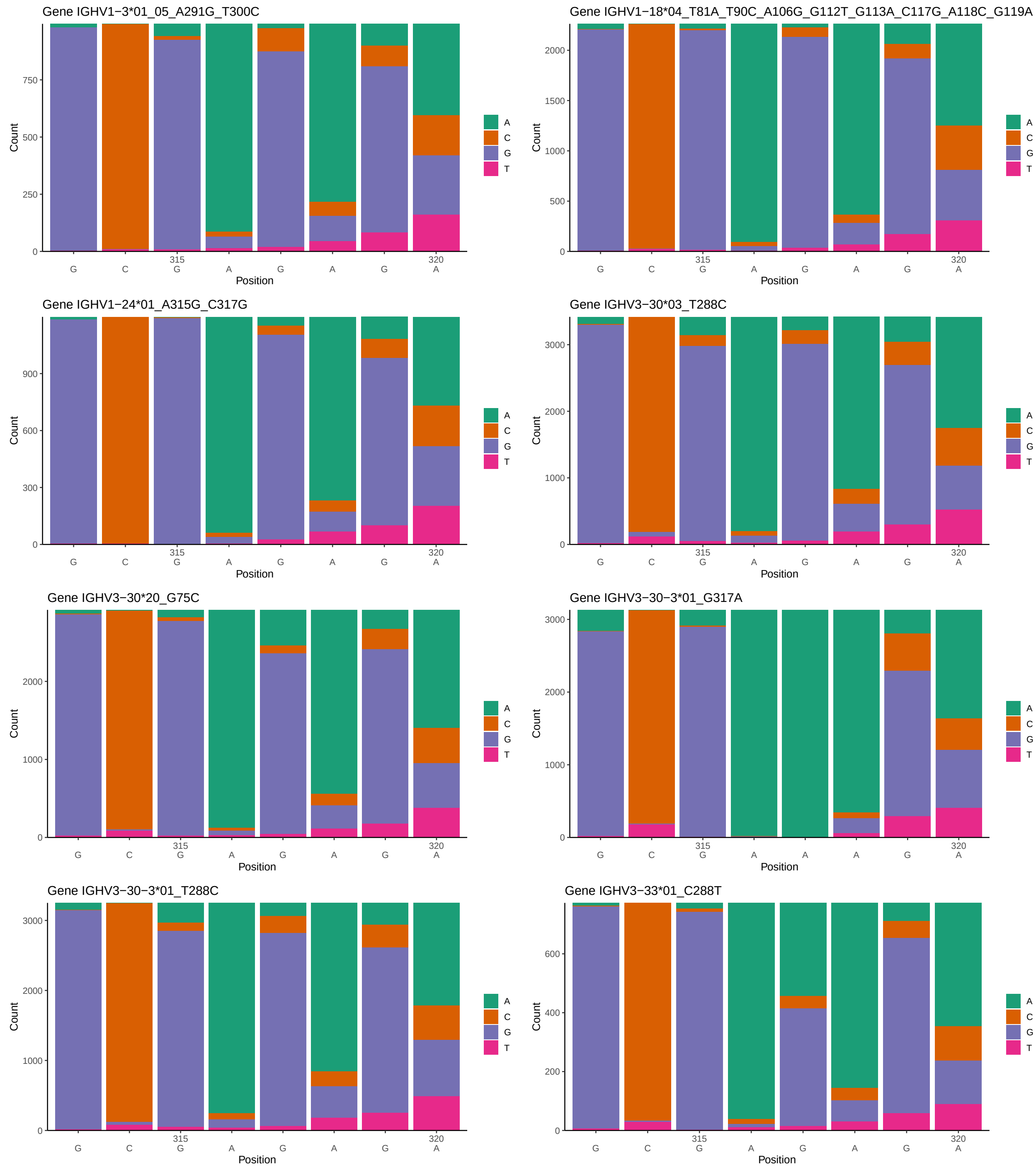
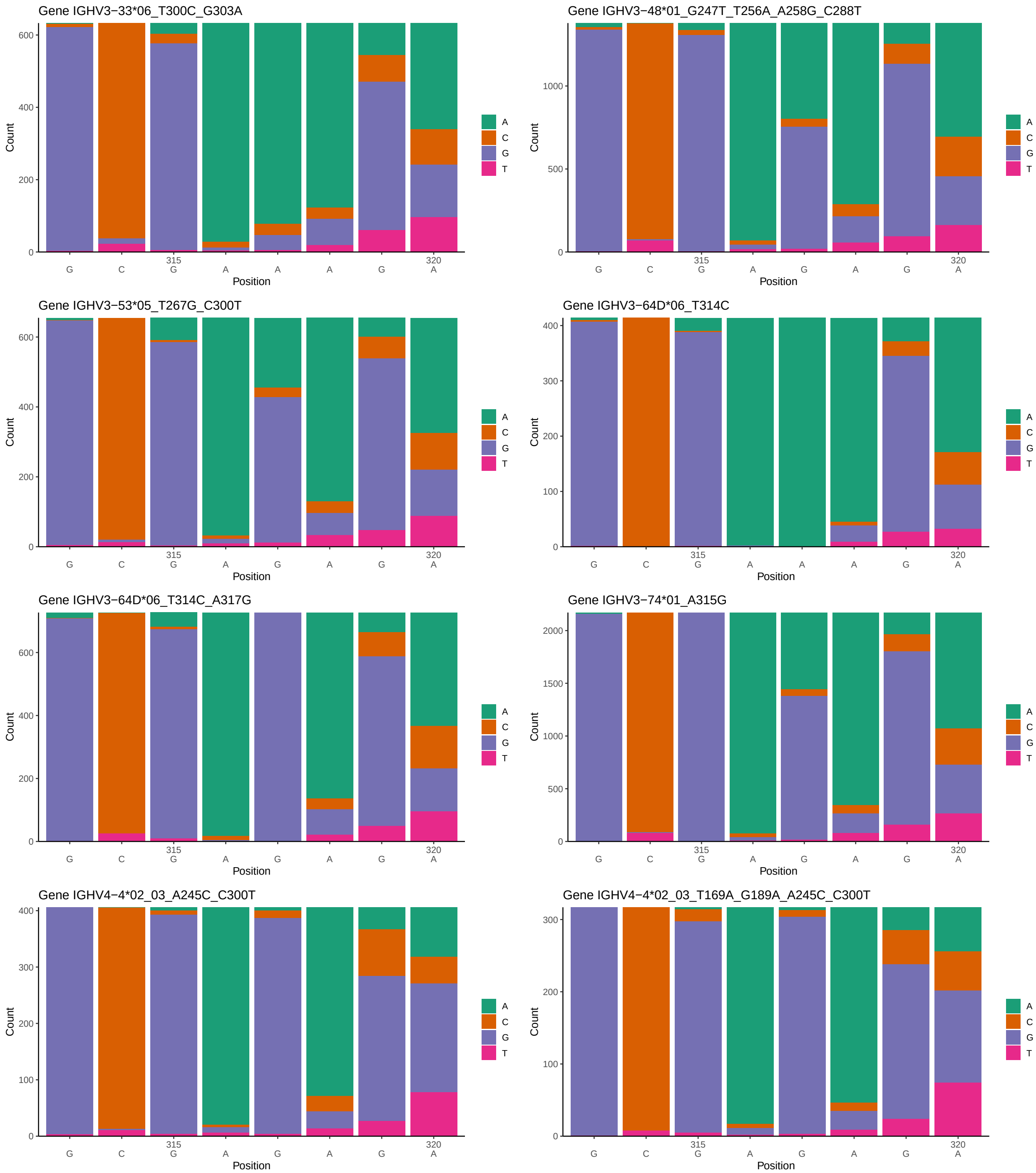
OGRDBstats Report

Contents

1 Novel sequence analysis

1.1 End-nucleotide composition





Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*01_G247T_T256A_A258G_C288T
Gene IGHV3−53*05_T267G_C300T
Gene IGHV3−64D*06_T314C
Gene IGHV3−64D*06_T314C_A317G
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−4*02_03_T169A_G189A_A245C_C300T
Count
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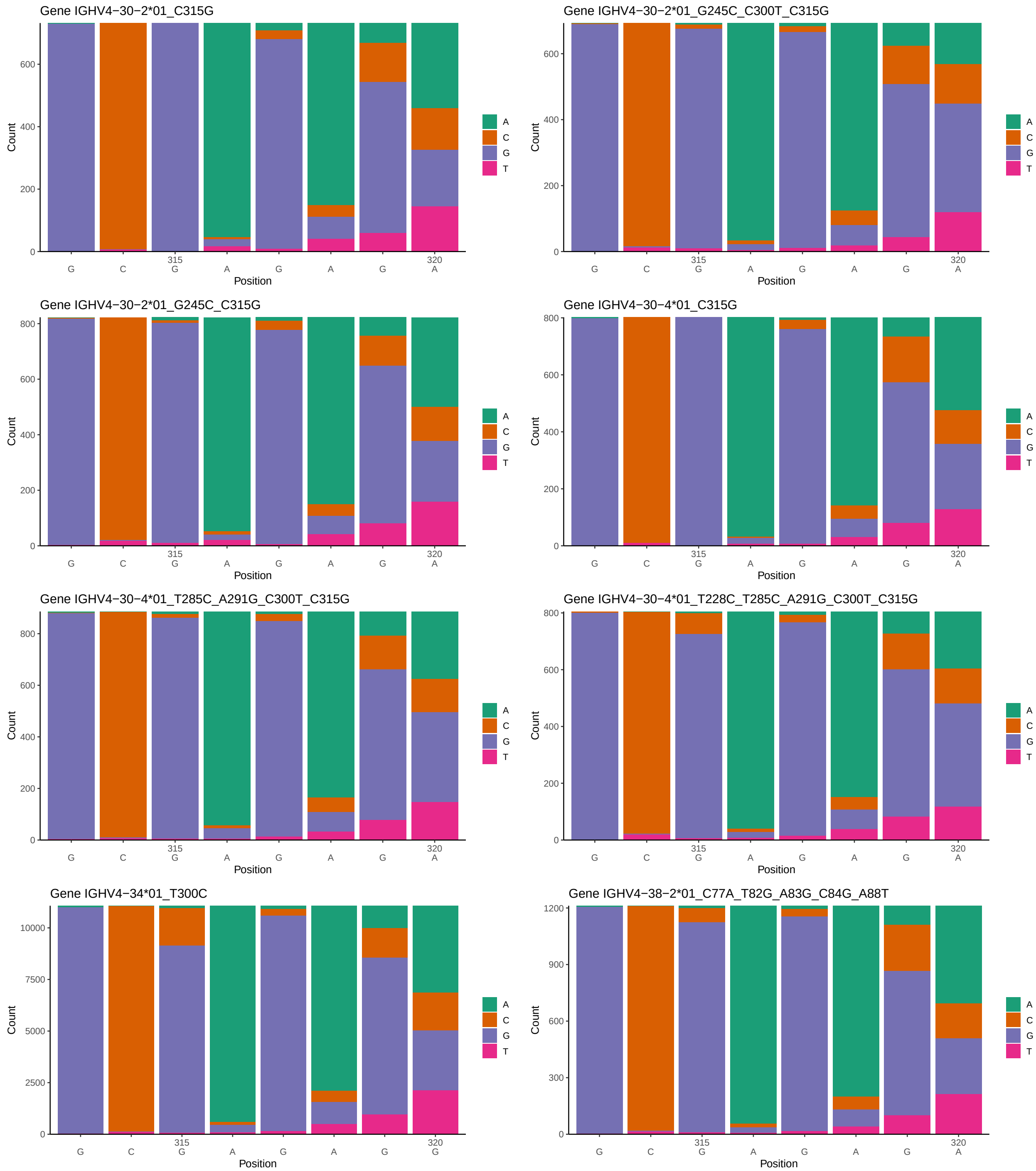
G C G A G A G A
A
C
G
T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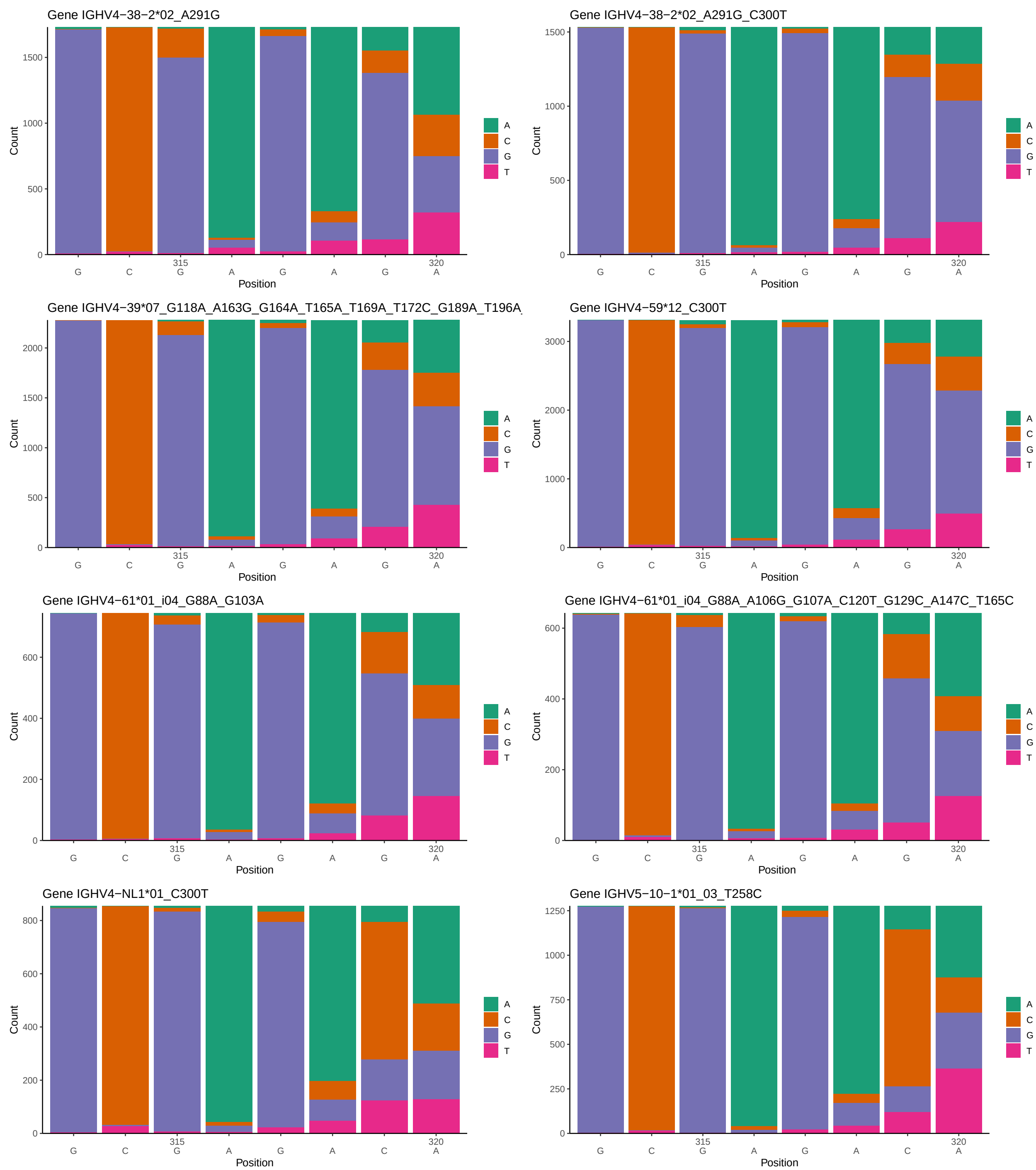
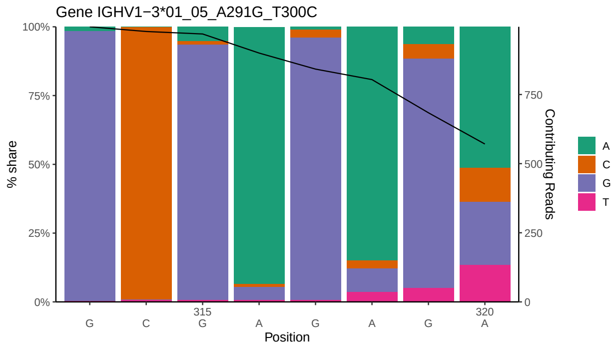
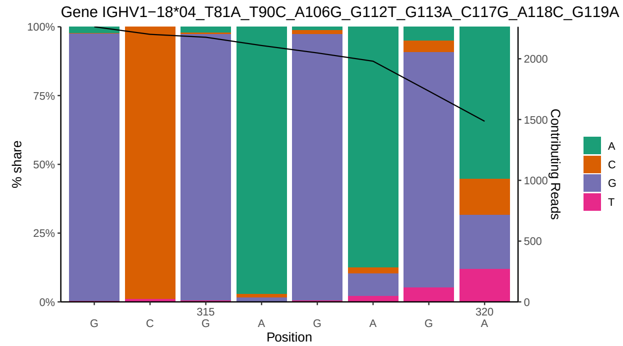
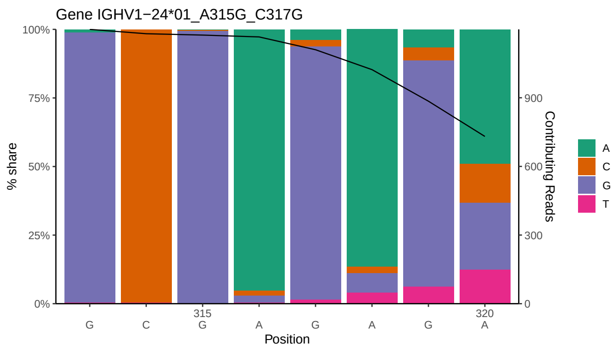
Gene IGHV4−30−4*01_C315G
Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
Gene IGHV4−30−4*01_T228C_T285C_A291G_C300T_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*01_C77A_T82G_A83G_C84G_A88T
Count
Position
A
C
G
T

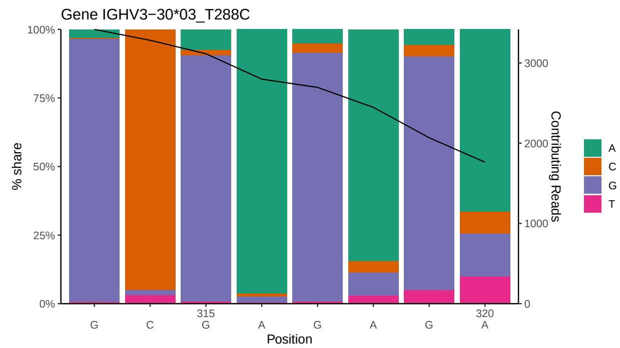

Gene IGHV4−38−2*02_A291G
Gene IGHV4−38−2*02_A291G_C300T
Gene IGHV4−39*07_G118A_A163G_G164A_T165A_T169A_T172C_G189A_T196A
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_G88A_G103A
Gene IGHV4−61*01_i04_G88A_A106G_G107A_C120T_G129C_A147C_T165C
Gene IGHV4−NL1*01_C300T
Gene IGHV5−10−1*01_03_T258C
Count
Position
A
C
G
T

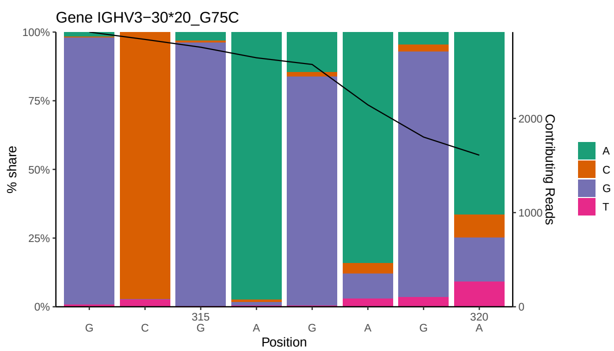
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

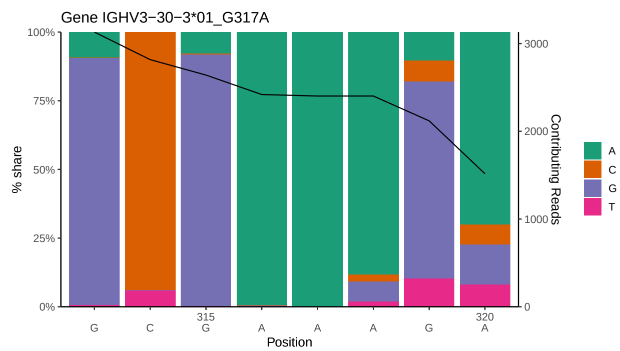


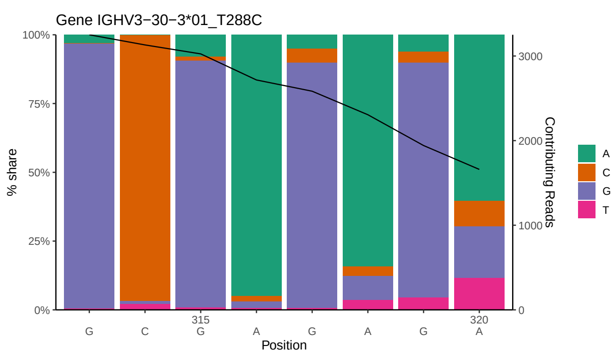


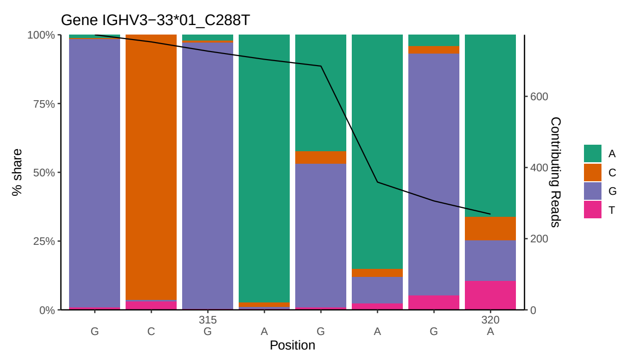




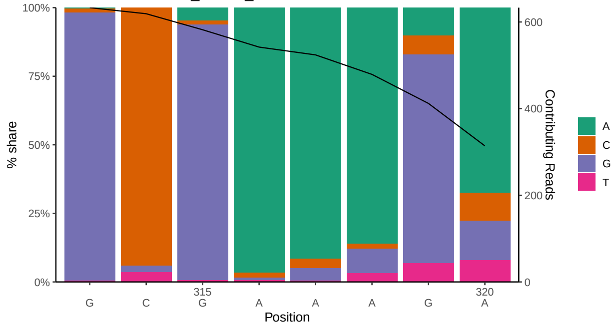








Gene IGHV3−33*06_T300C_G303A



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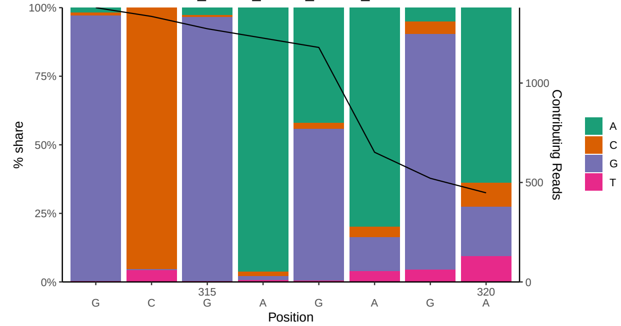

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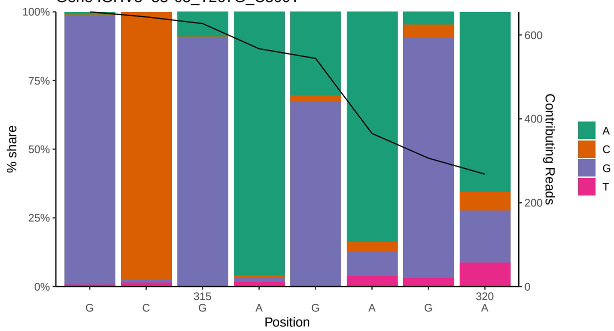

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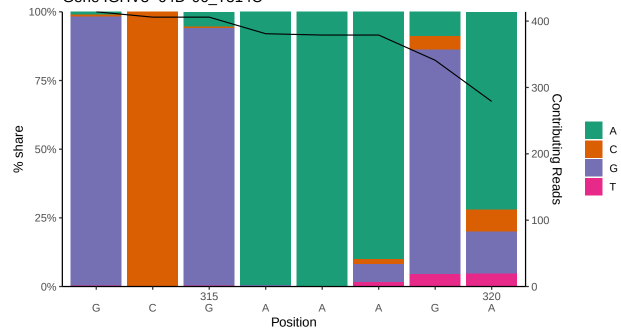

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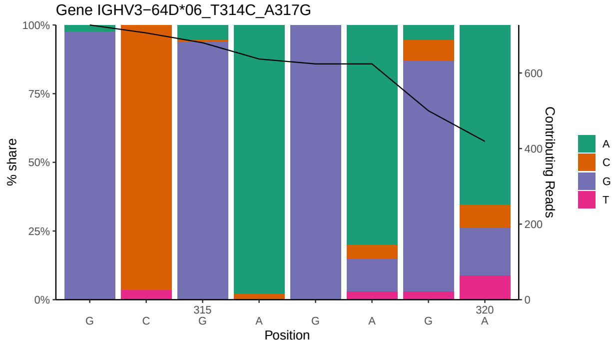

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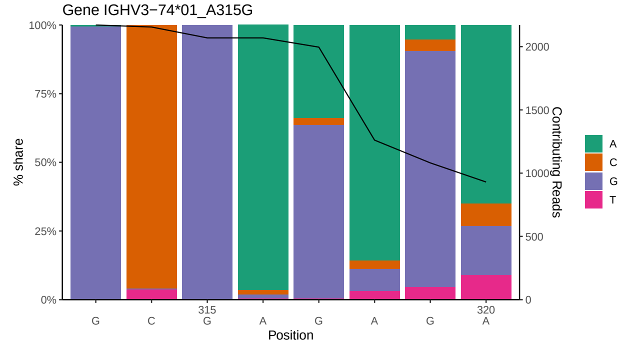

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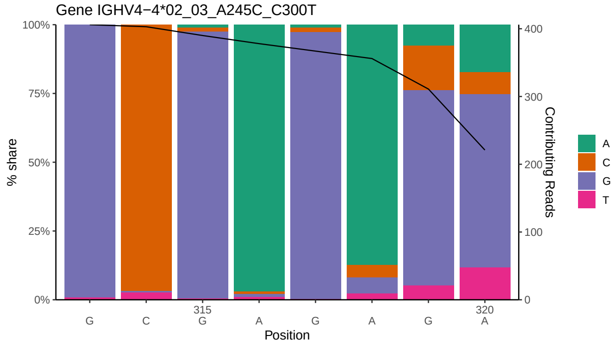

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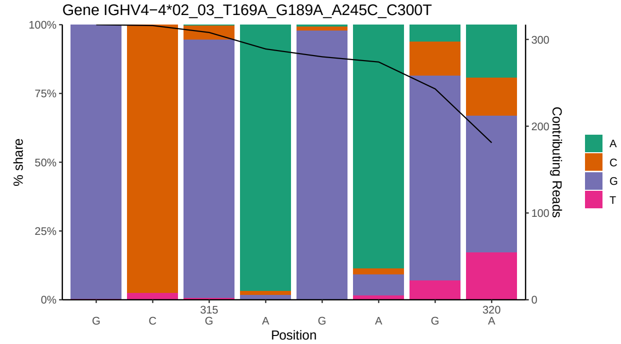

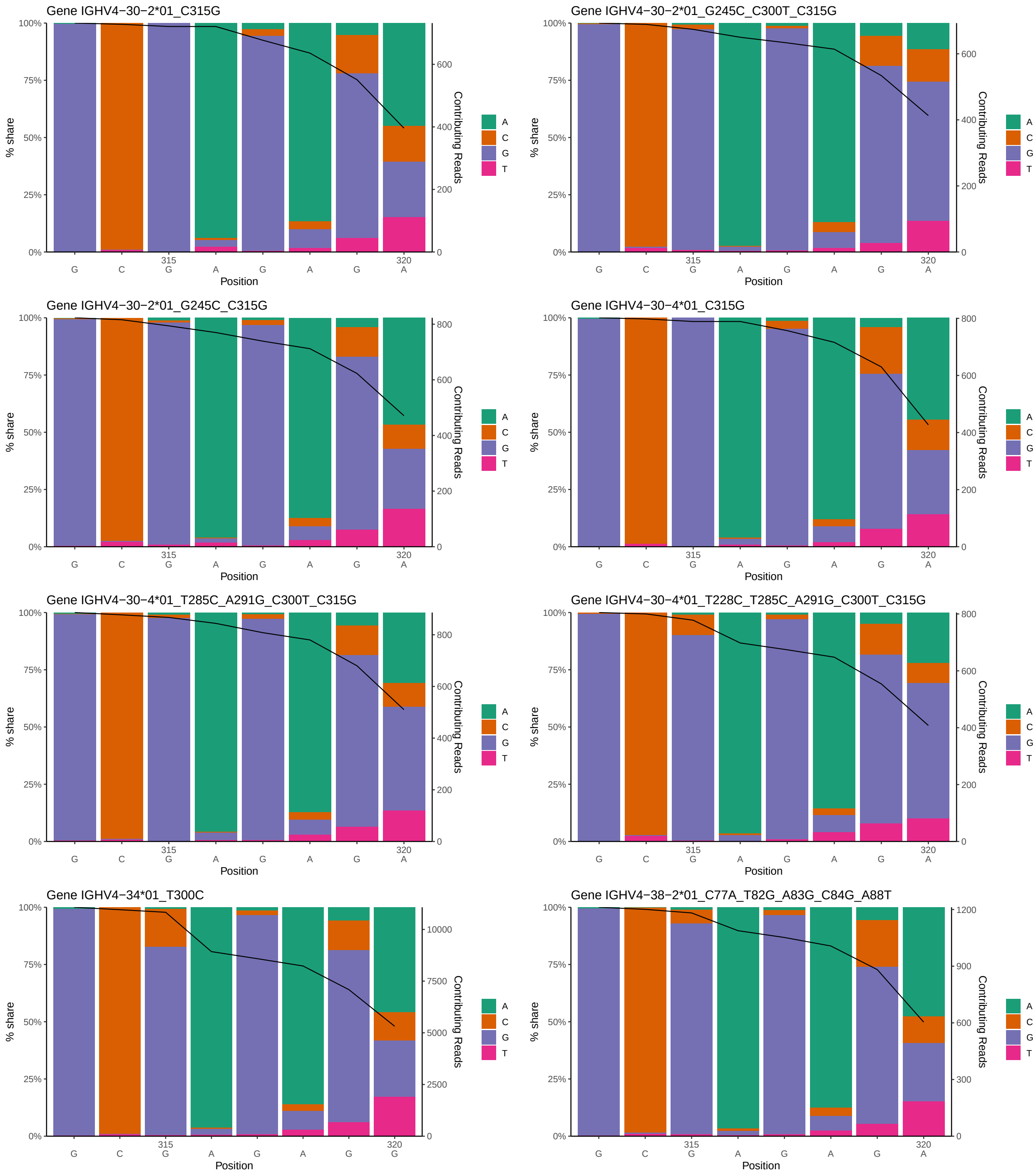
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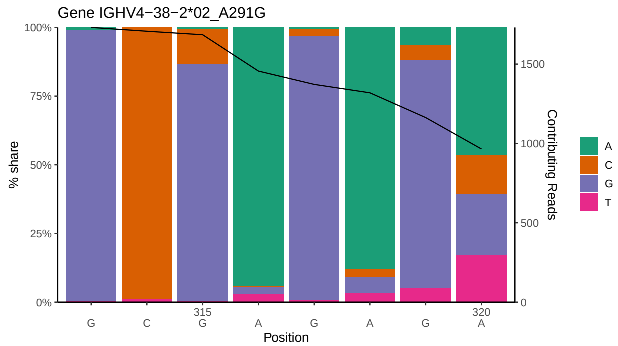
Gene IGHV4−30−4*01_C315G
Gene IGHV4−30−4*01_T285C_A291G_C300T_C315G
Gene IGHV4−30−4*01_T228C_T285C_A291G_C300T_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*01_C77A_T82G_A83G_C84G_A88T
% share
Contributing Reads
Position
A
C
G
T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% share
Contributing Reads
Position
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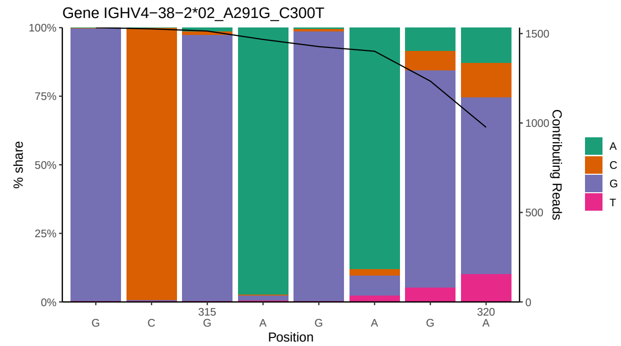
C
G
T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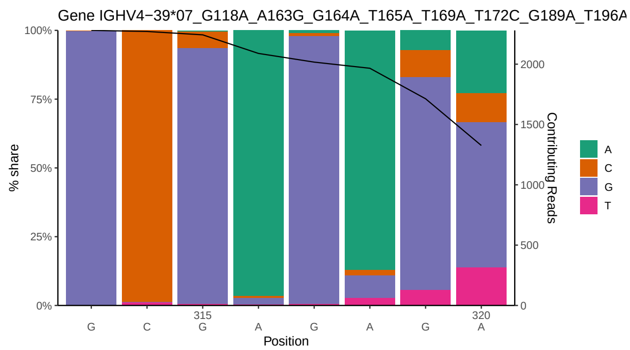
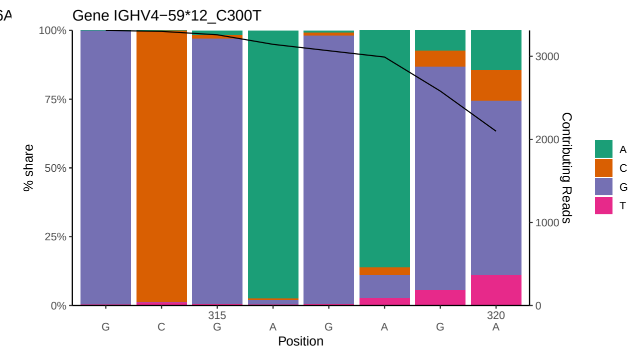
% share
Contributing Reads
Position
A
C
G
T


Gene IGHV4−39*07_G118A_A163G_G164A_T165A_T169A_T172C_G189A_T196A
% share
Contributing Reads
Position
A
C
G
T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Contributing Reads
Position
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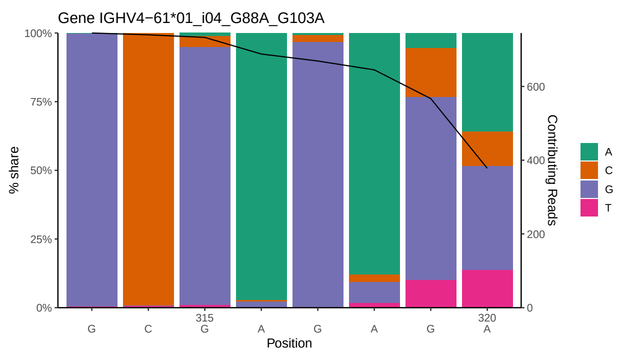
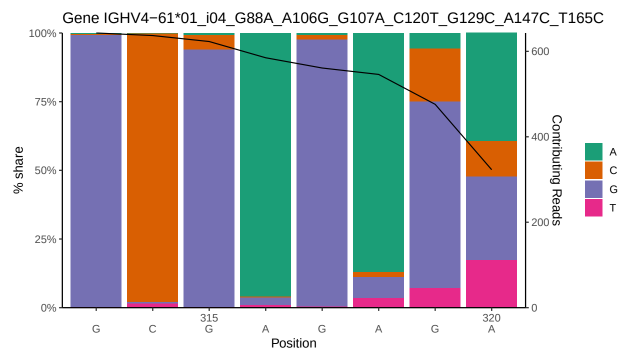
C
G
T


Gene IGHV4−61*01_i04_G88A_G103A
% share
Contributing Reads
Position
A
C
G
T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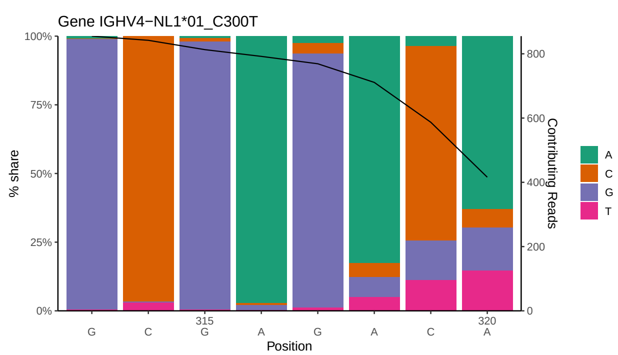
Contributing Reads
Position
A
C
G
T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Contributing Reads
Position
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C
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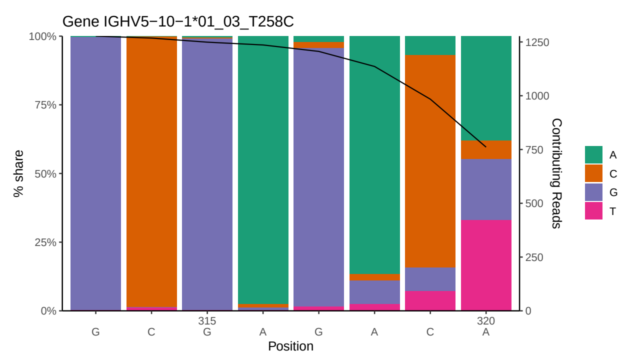
Gene IGHV5−10−1*01_03_T258C
% share
Contributing Reads
Position
A
C
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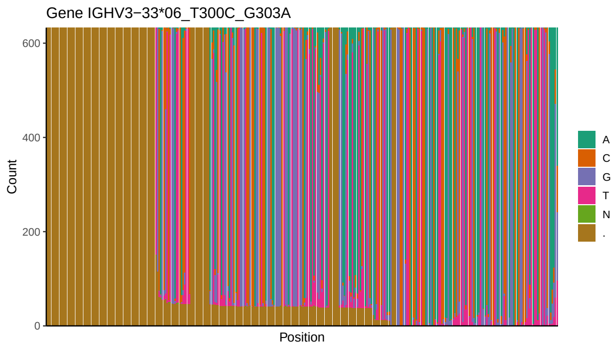
T

1.3 Whole-sequence composition of each assigned read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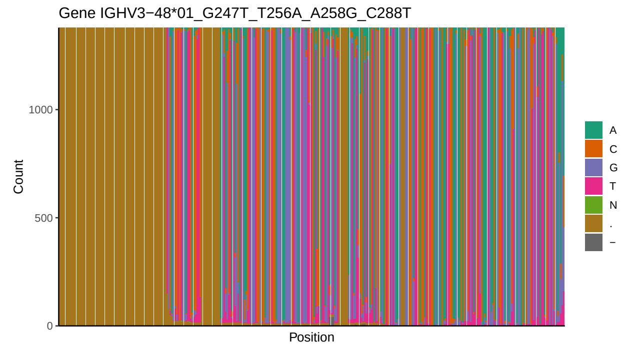
Position
A
C
G
T
N
.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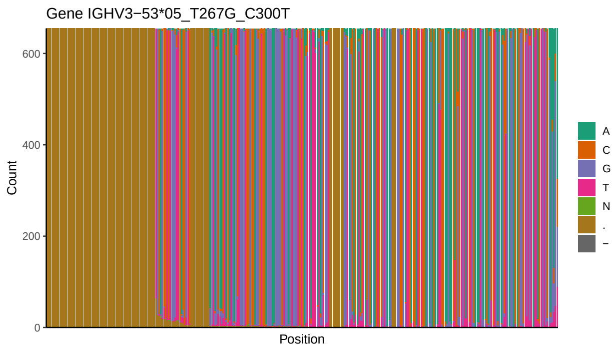
Count
1000
500
0
Position
A
C
G
T
N
.
−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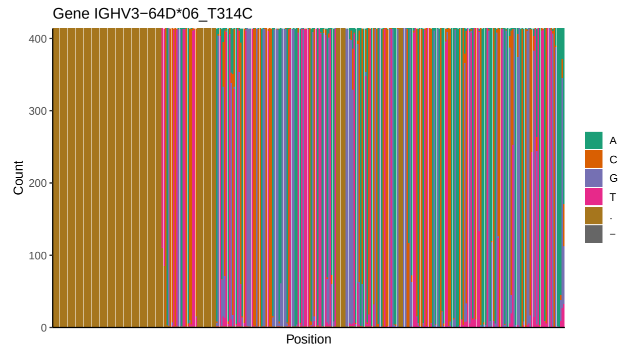
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C
G
T
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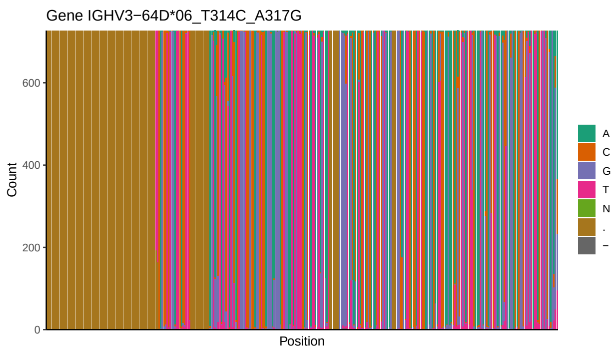
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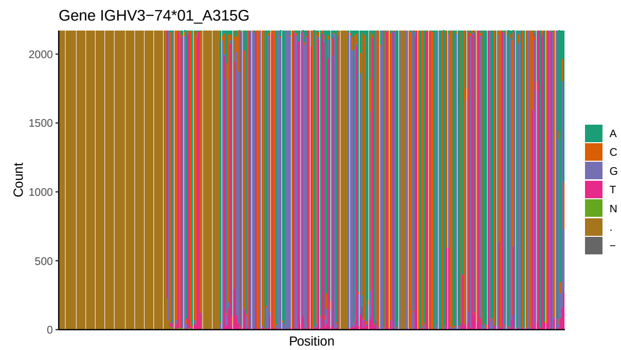
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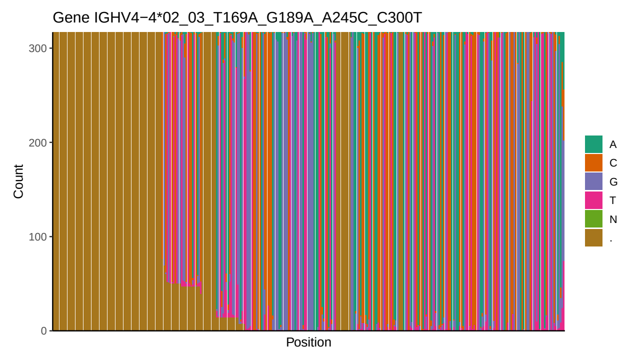
N
.
−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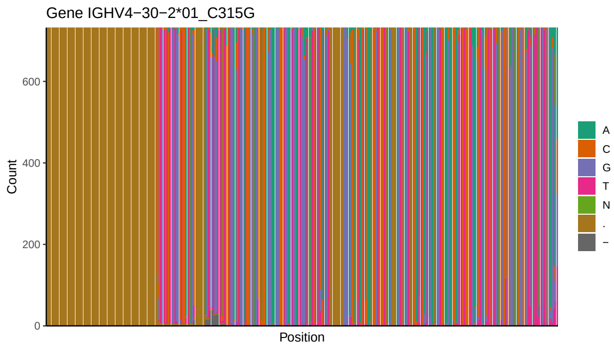
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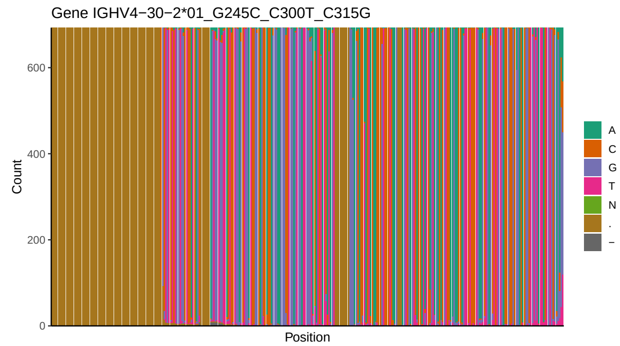
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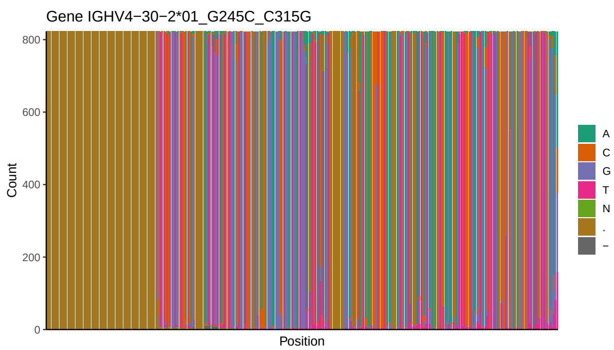
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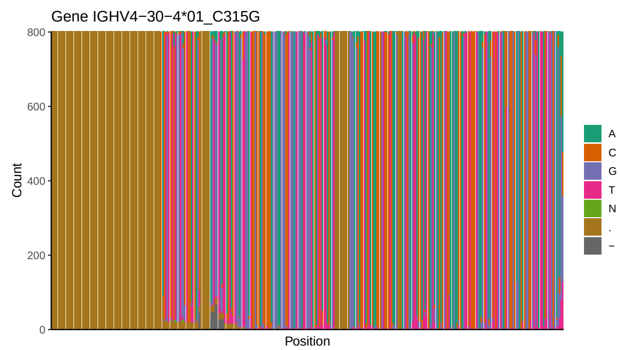
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C
G
T
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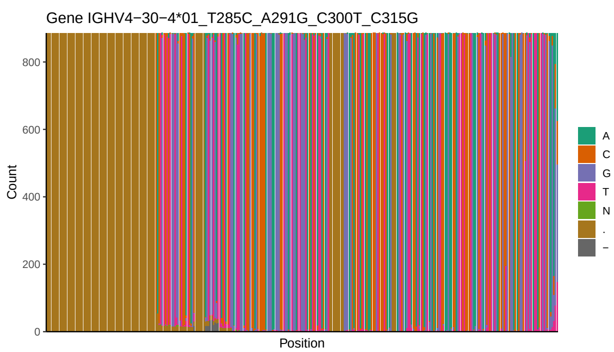
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Position
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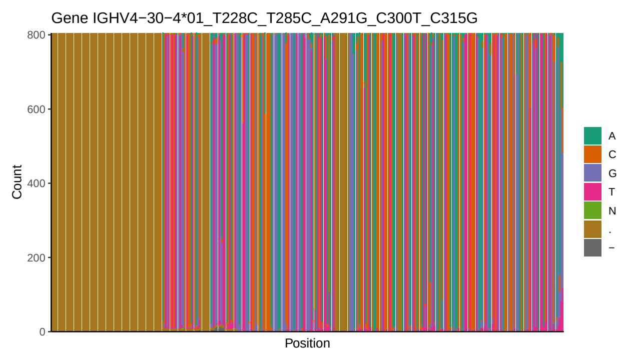
G
T
N
.
−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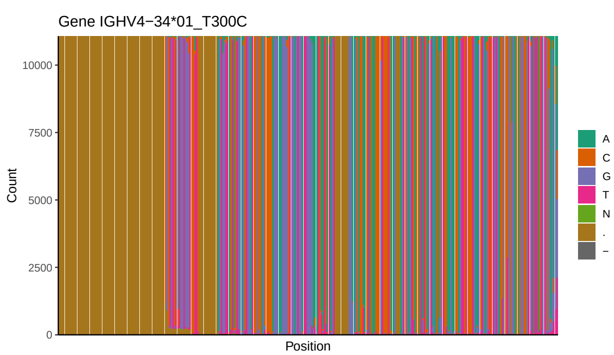
T
N
.
−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7500
10000
Position
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C
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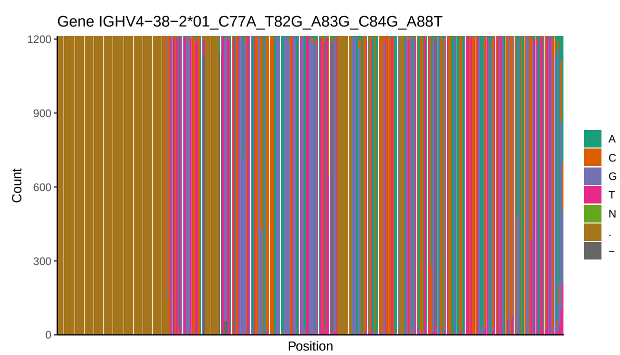
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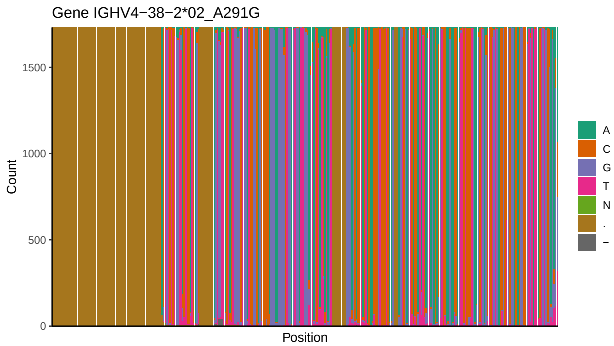
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C
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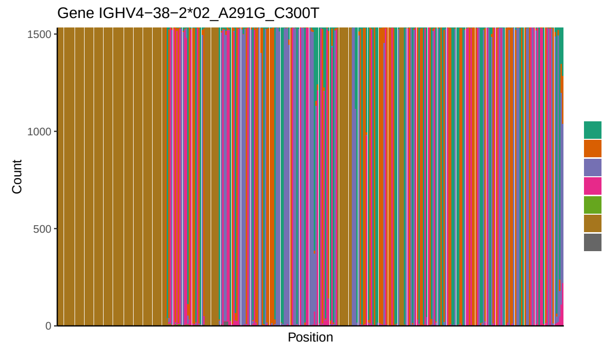
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Position
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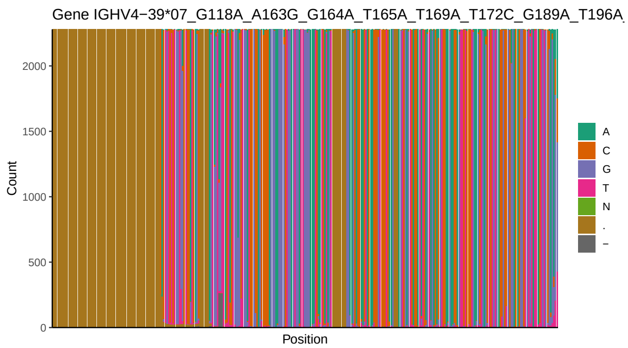
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Position
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C
G
T
N
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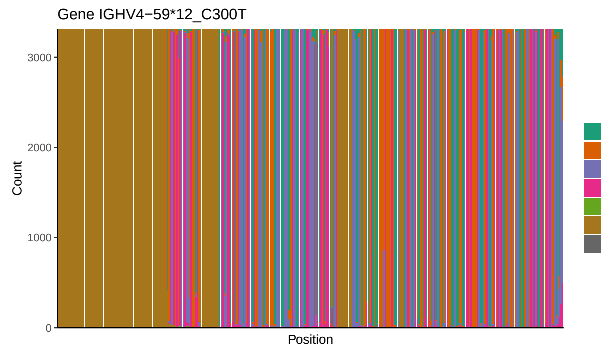
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Count
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1000
2000
3000
Position
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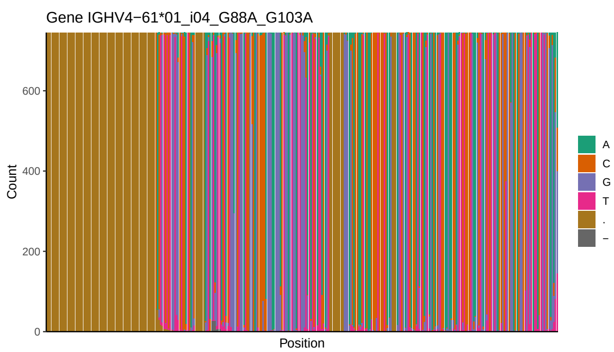
C
G
T
N
.
−


Gene IGHV4−61*01_i04_G88A_G103A
Count
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200
400
600
Position
A
C
G
T
.
−


Gene IGHV4−61*01_i04_G88A_A106G_G107A_C120T_G129C_A147C_T165C
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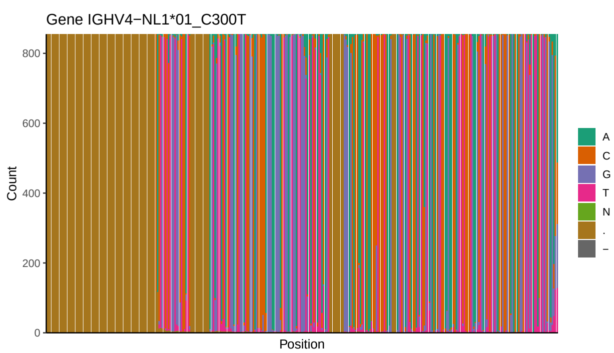
200
400
600
Position
A
C
G
T
N
.
−


Gene IGHV4−NL1*01_C300T
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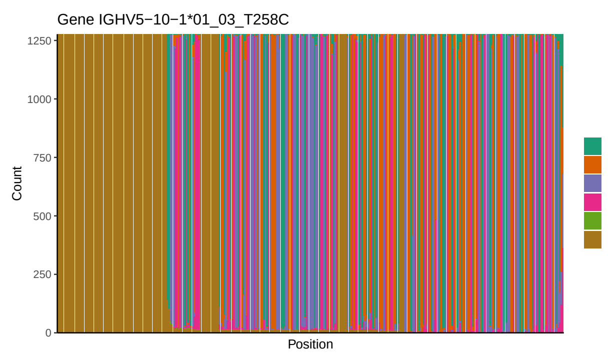
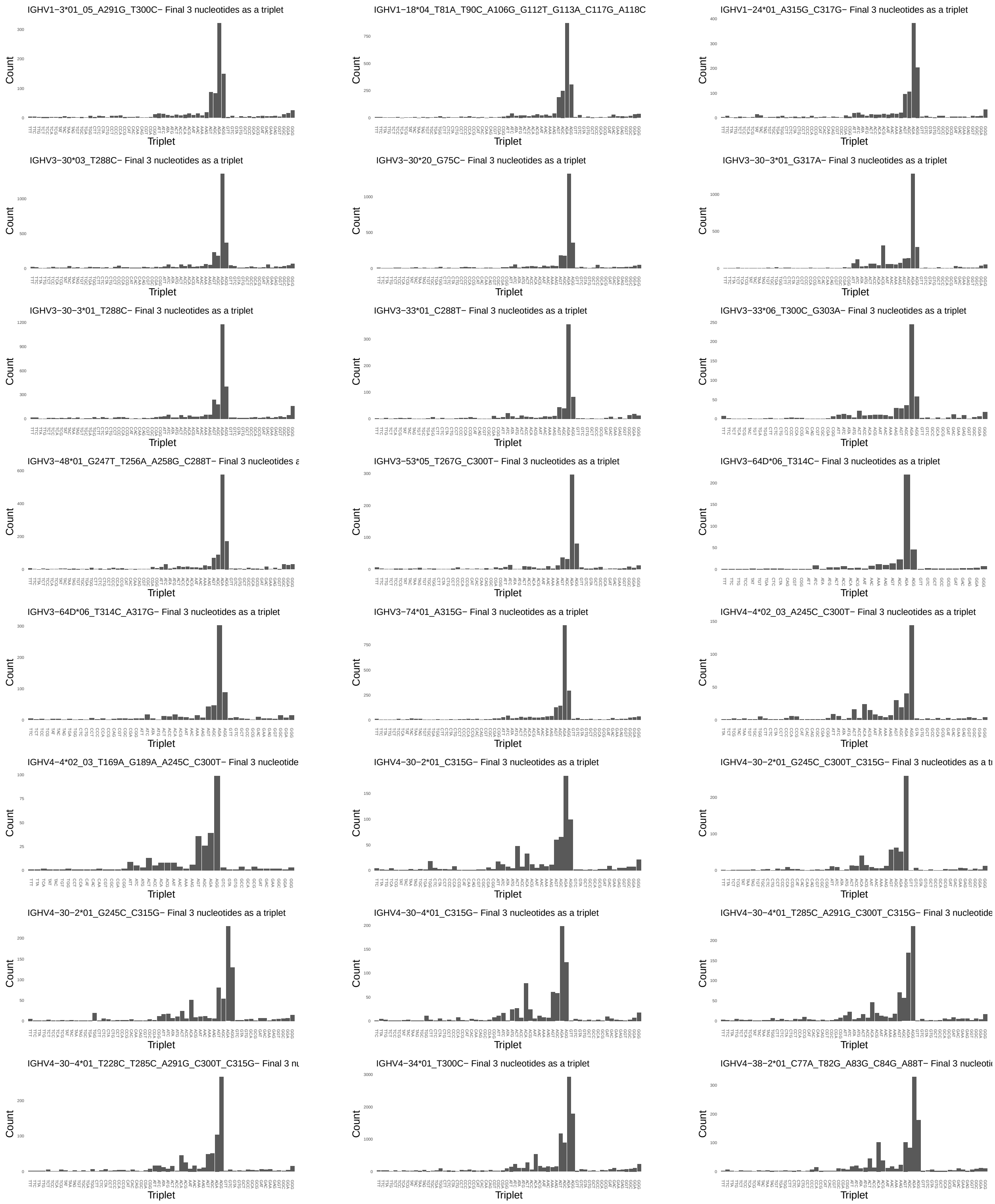
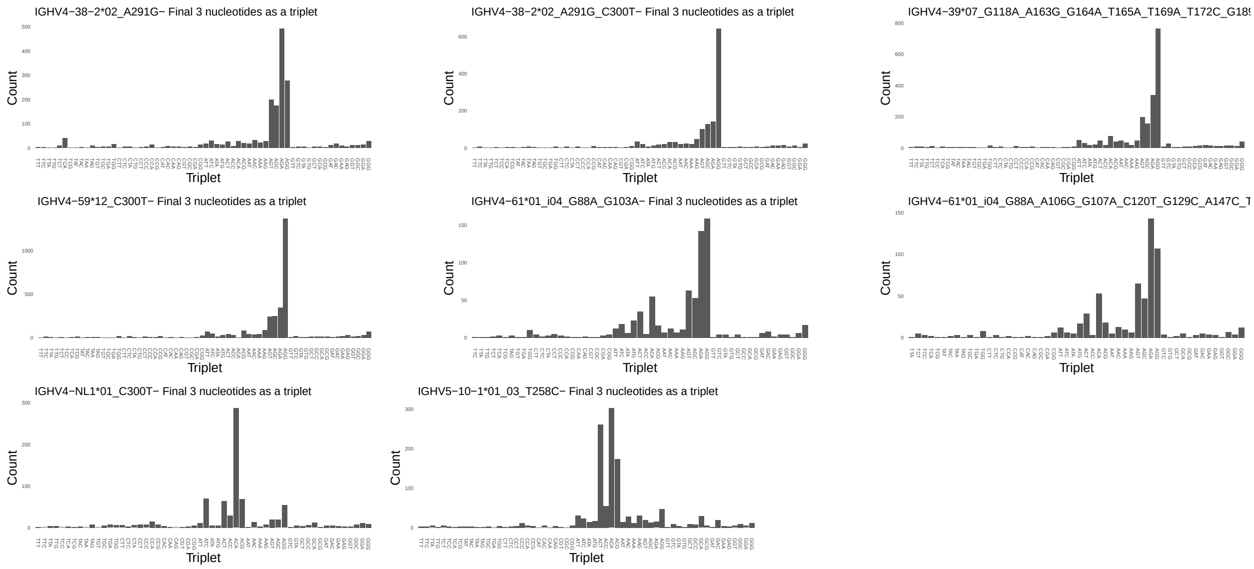
200
400
600
800
Position
A
C
G
T
N
.
−

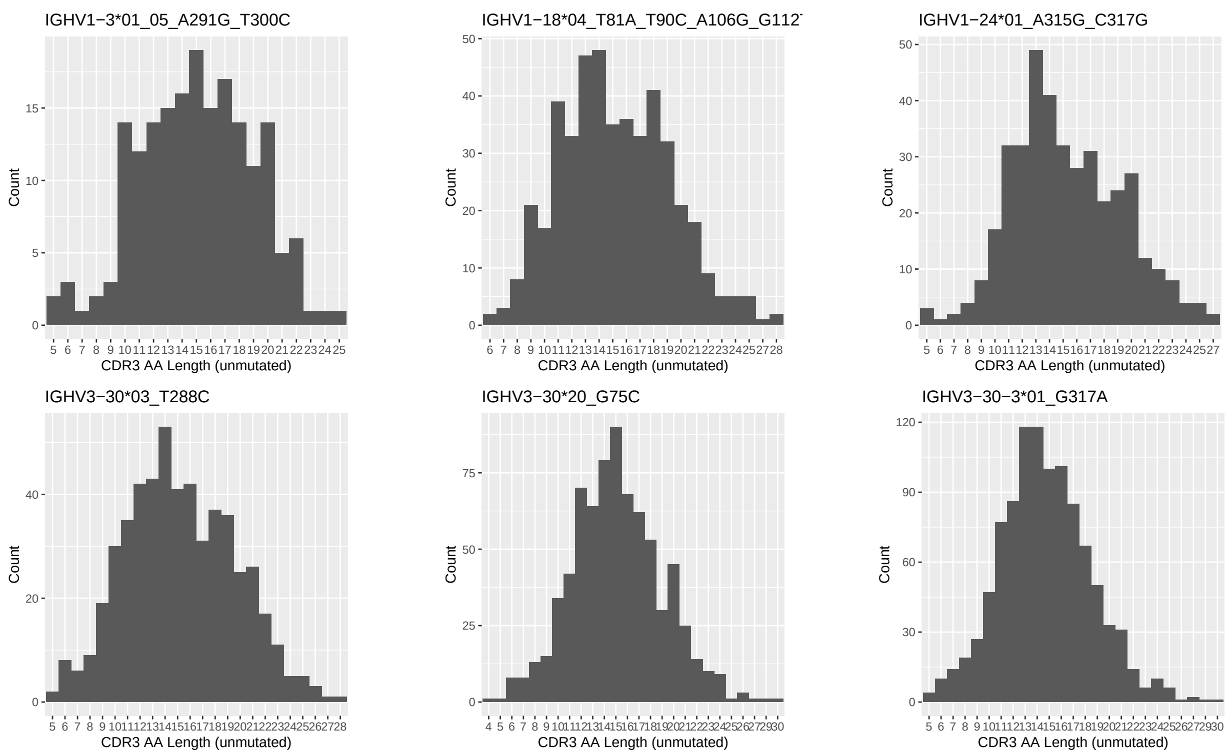

Gene IGHV5−10−1*01_03_T258C
Count
0
250
500
750
1000
1250
Position
A
C
G
T
N
.

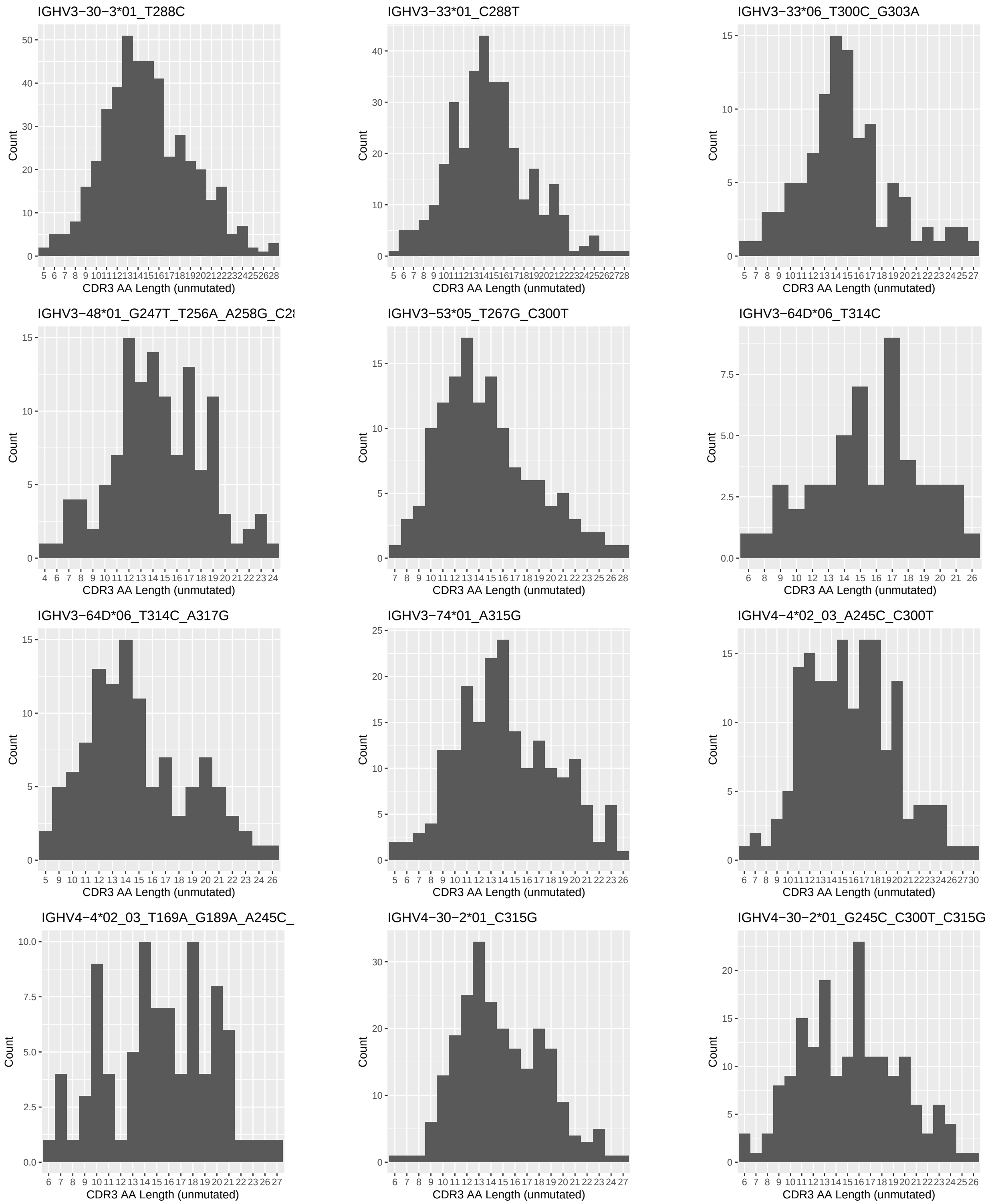
1.4 Final three nucleotides: frequency of each observed triplet

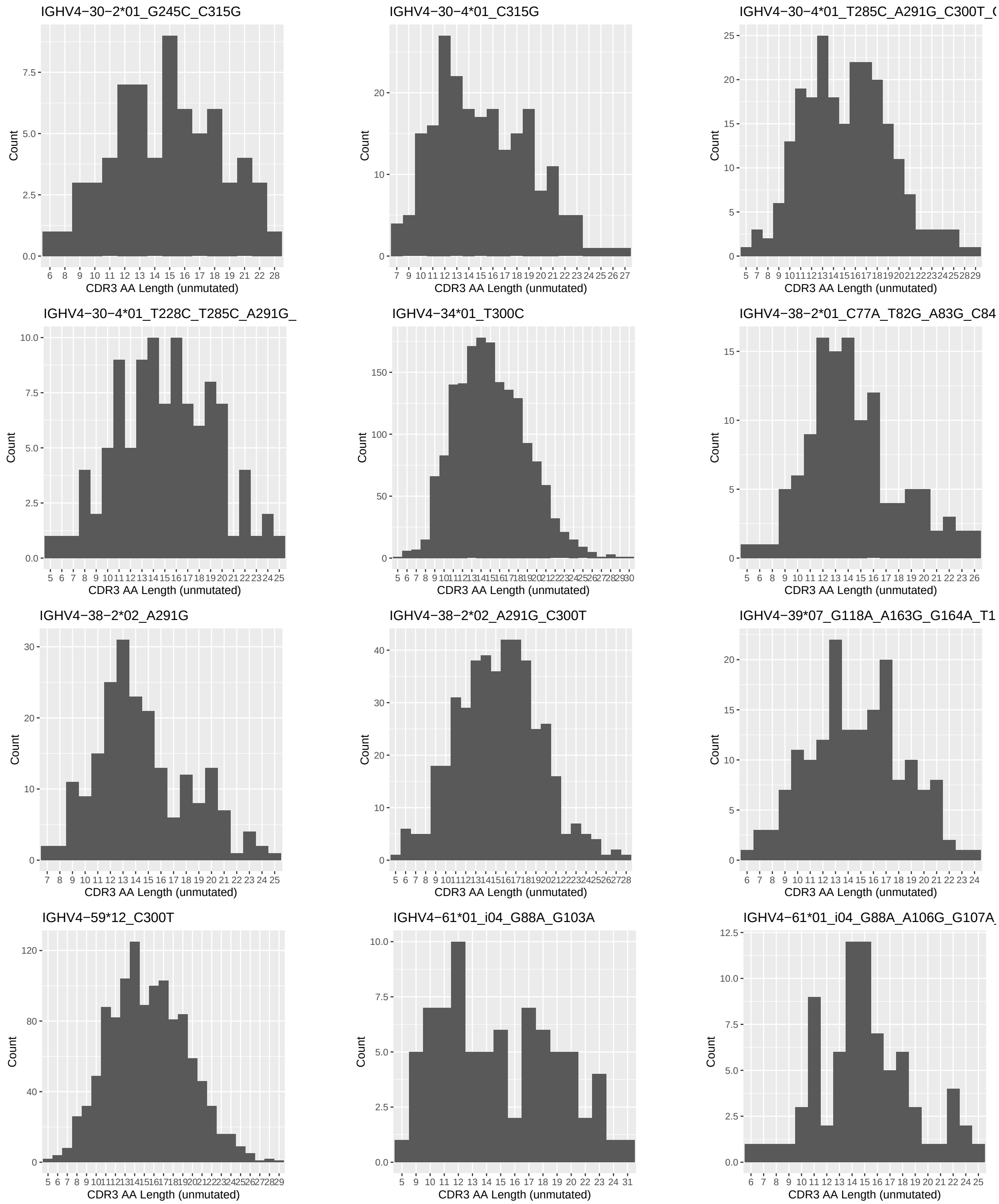


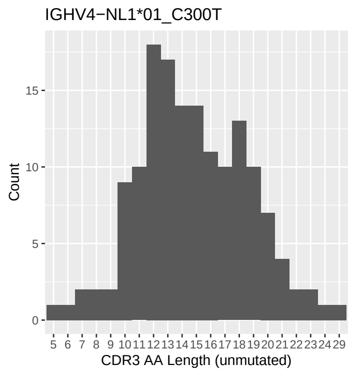


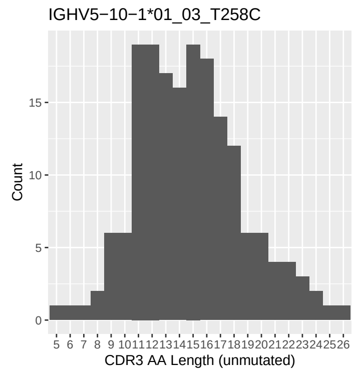
1.5 CDR3 length distribution, in assignments to novel alleles



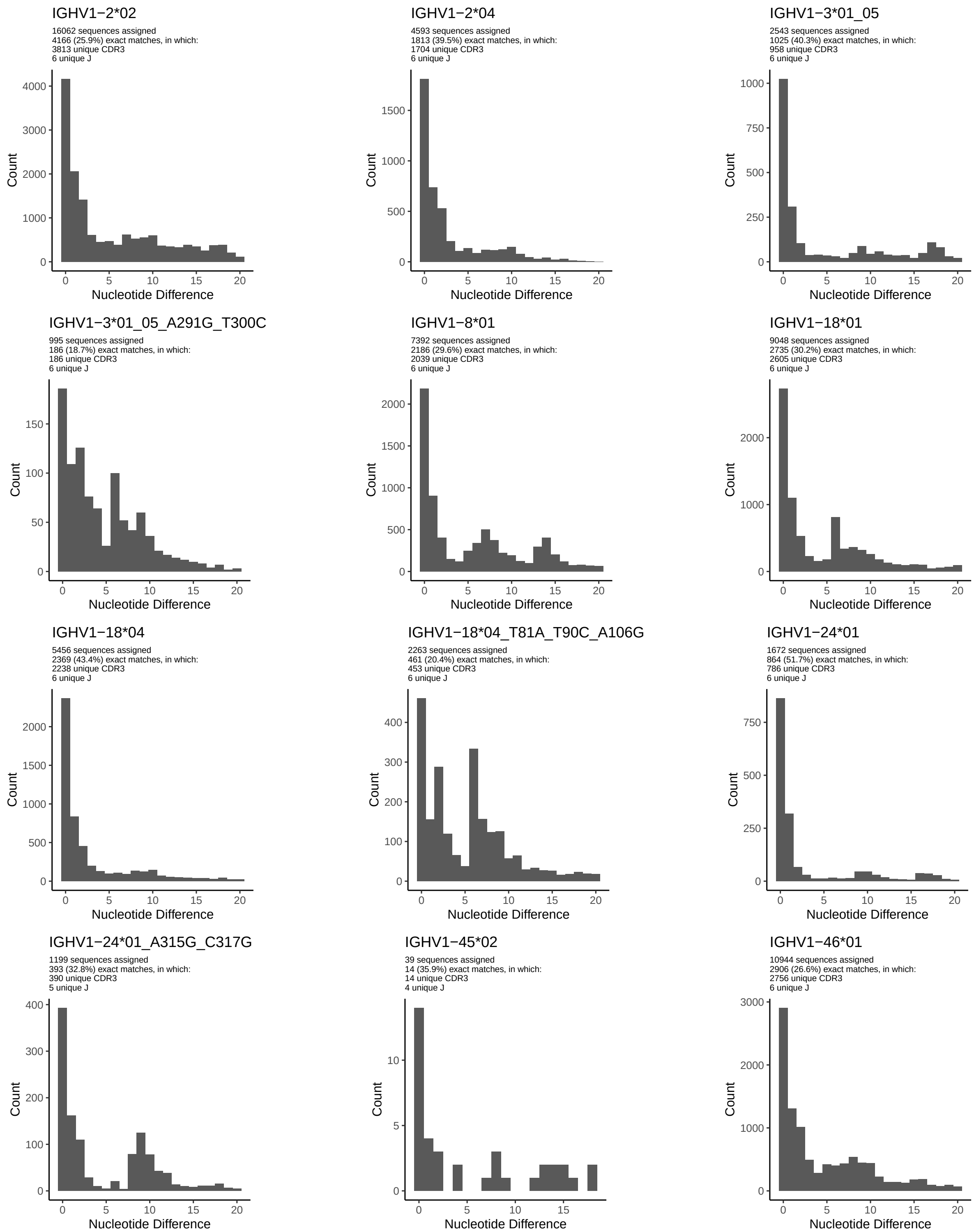

IGHV3−30−3*01_T288C
IGHV3−33*01_C288T
IGHV3−33*06_T300C_G303A
IGHV3−48*01_G247T_T256A_A258G_C2
IGHV3−53*05_T267G_C300T
IGHV3−64D*06_T314C
IGHV3−64D*06_T314C_A317G
IGHV3−74*01_A315G
IGHV4−4*02_03_A245C_C300T
IGHV4−4*02_03_T169A_G189A_A245C_
IGHV4−30−2*01_C315G
IGHV4−30−2*01_G245C_C300T_C315G
Count
CDR3 AA Length (unmutated)

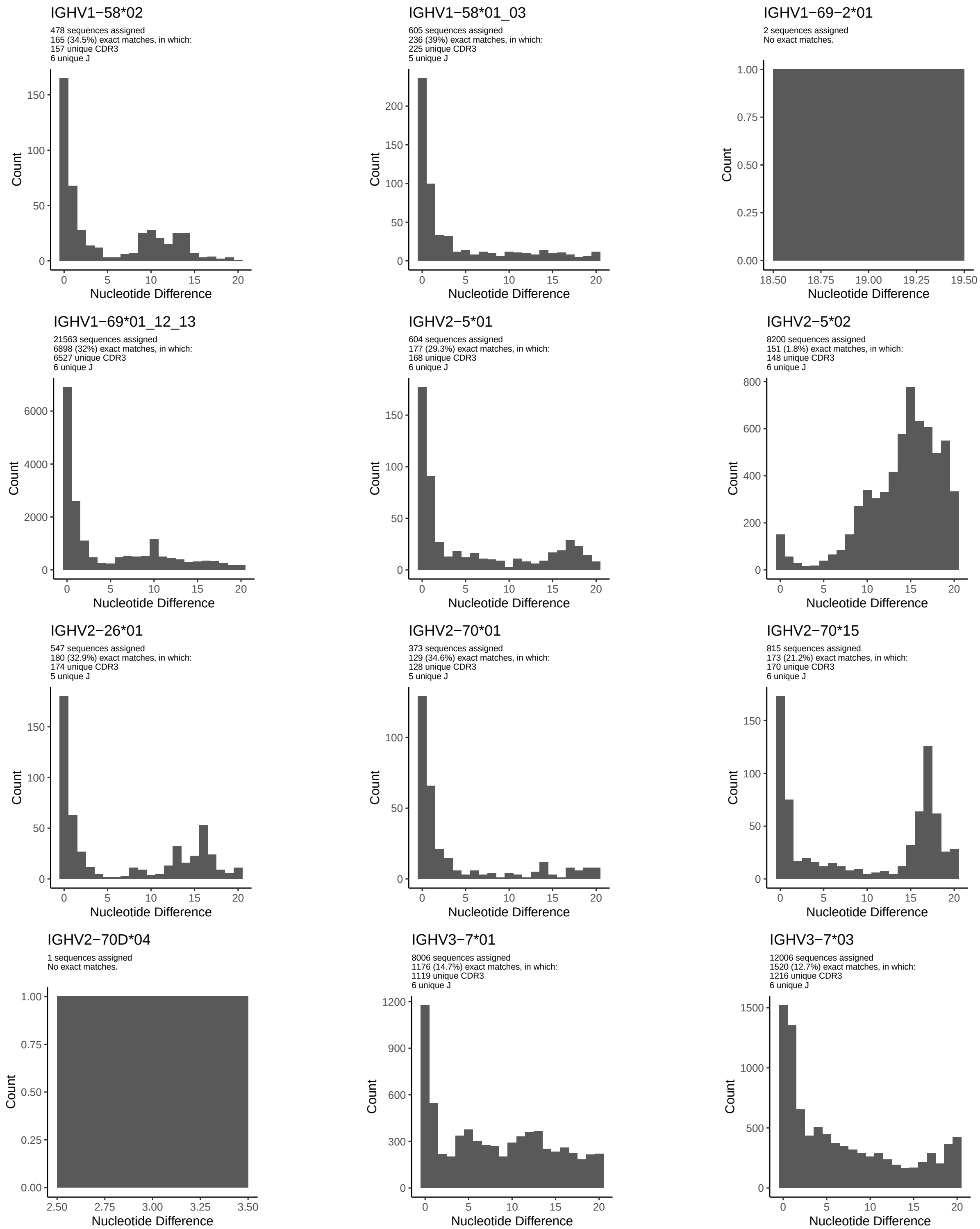

IGHV4−30−2*01_G245C_C315G
IGHV4−30−4*01_C315G
IGHV4−30−4*01_T285C_A291G_C300T_(
IGHV4−30−4*01_T228C_T285C_A291G_
IGHV4−34*01_T300C
IGHV4−38−2*01_C77A_T82G_A83G_C84
IGHV4−38−2*02_A291G
IGHV4−38−2*02_A291G_C300T
IGHV4−39*07_G118A_A163G_G164A_T1
IGHV4−59*12_C300T
IGHV4−61*01_i04_G88A_G103A
IGHV4−61*01_i04_G88A_A106G_G107A_
Count
CDR3 AA Length (unmutated)

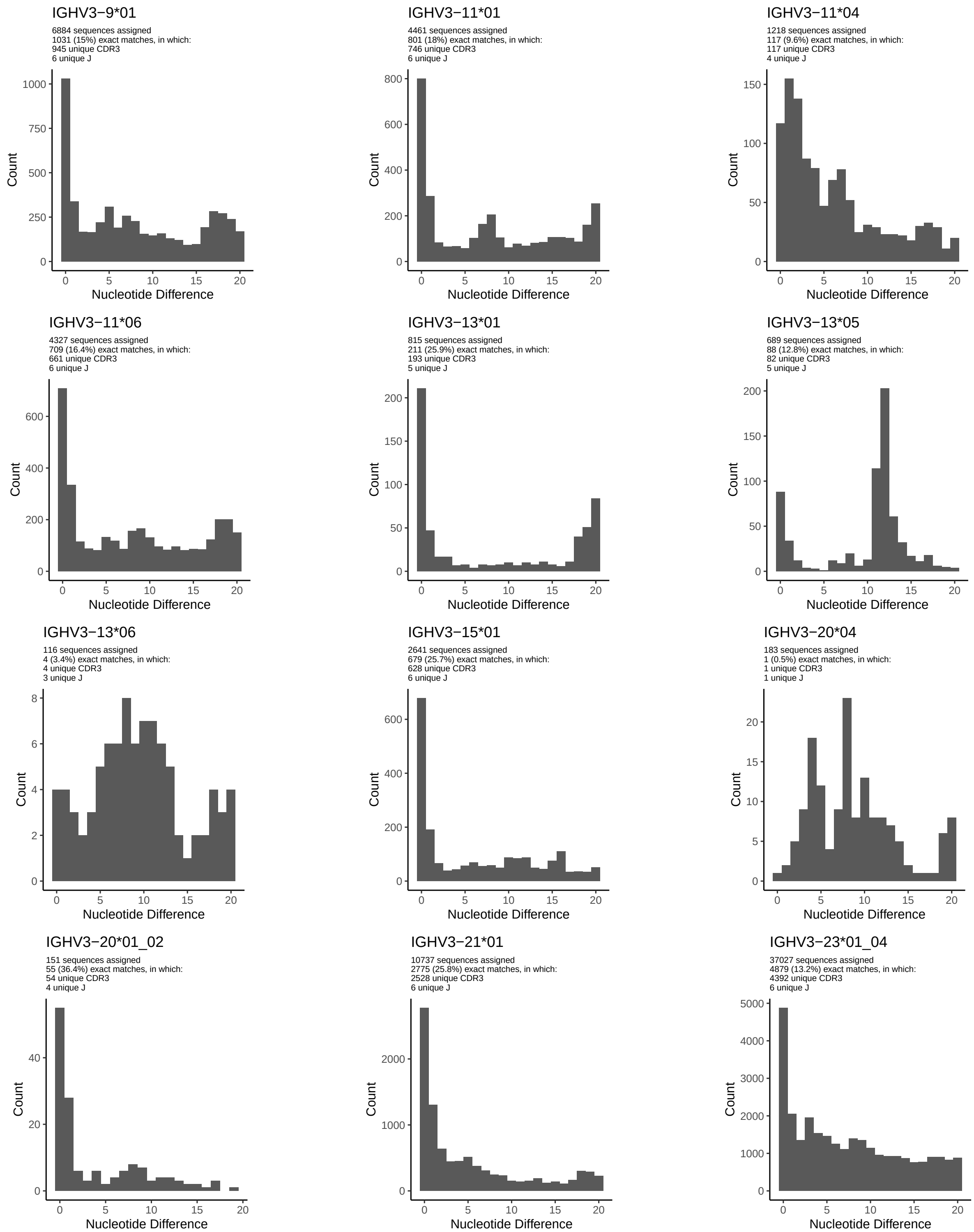

IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

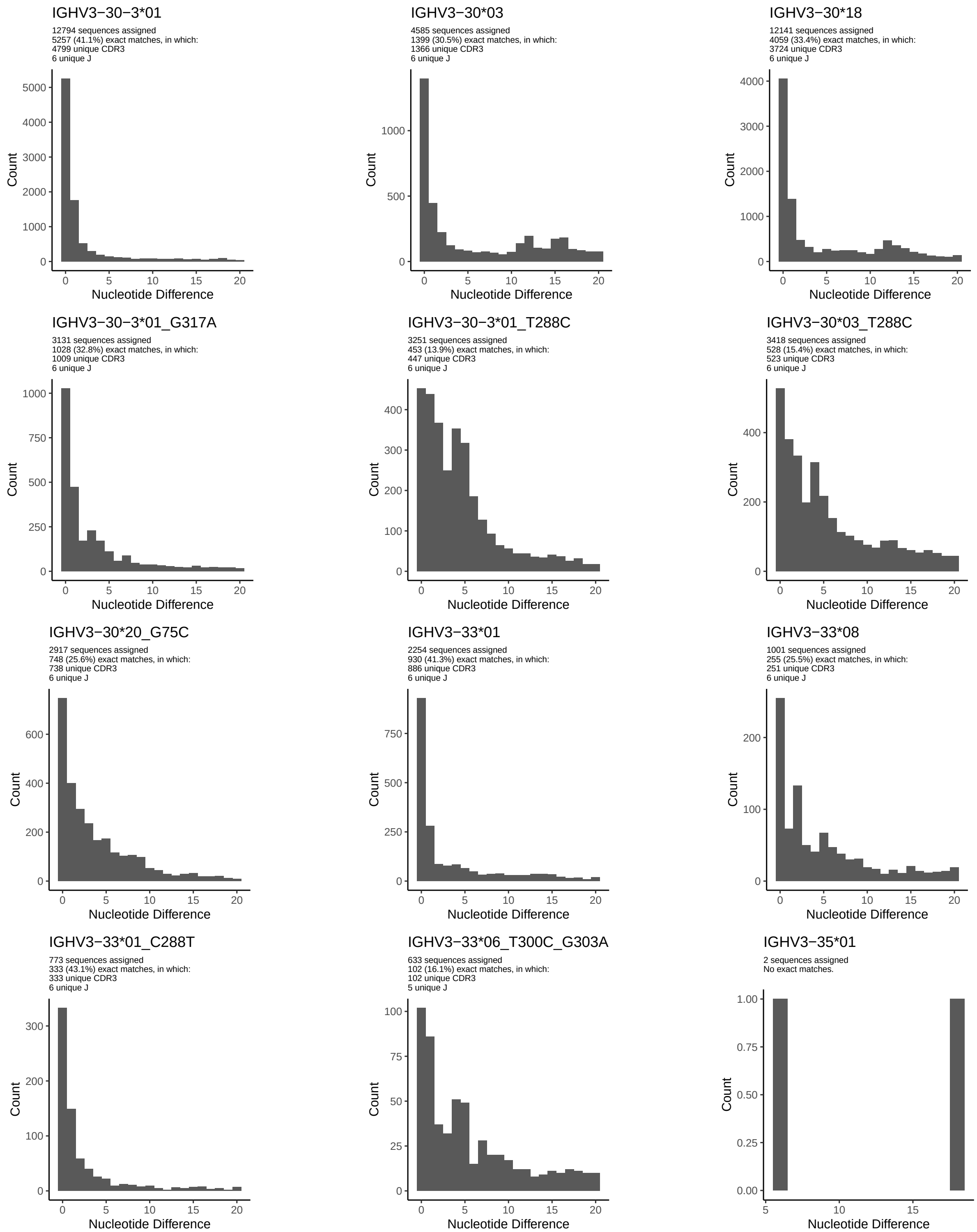

IGHV5−10−1*01_03_T258C
Count
CDR3 AA Length (unmutated)

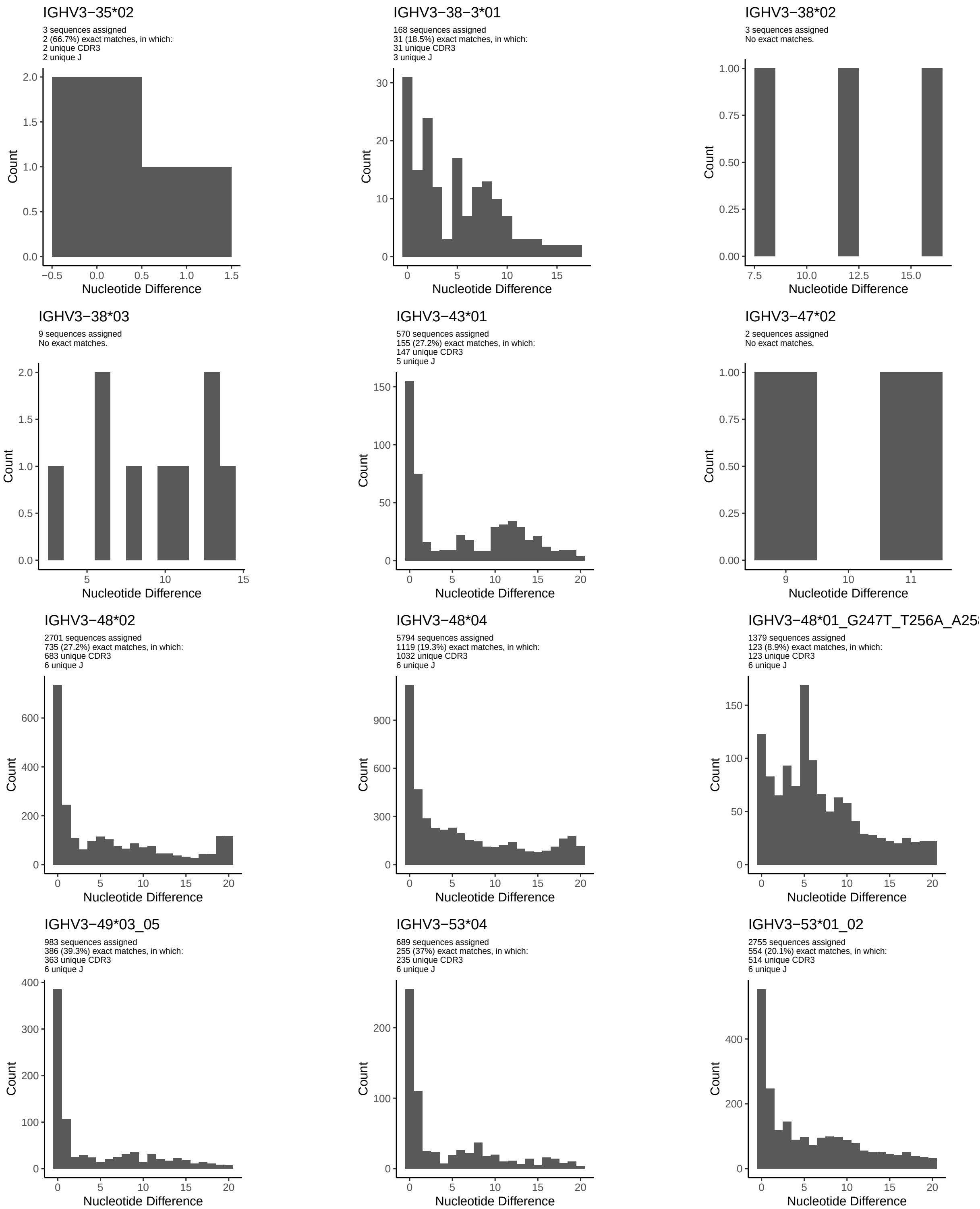
2 Variation from germline, in assignments to each allele

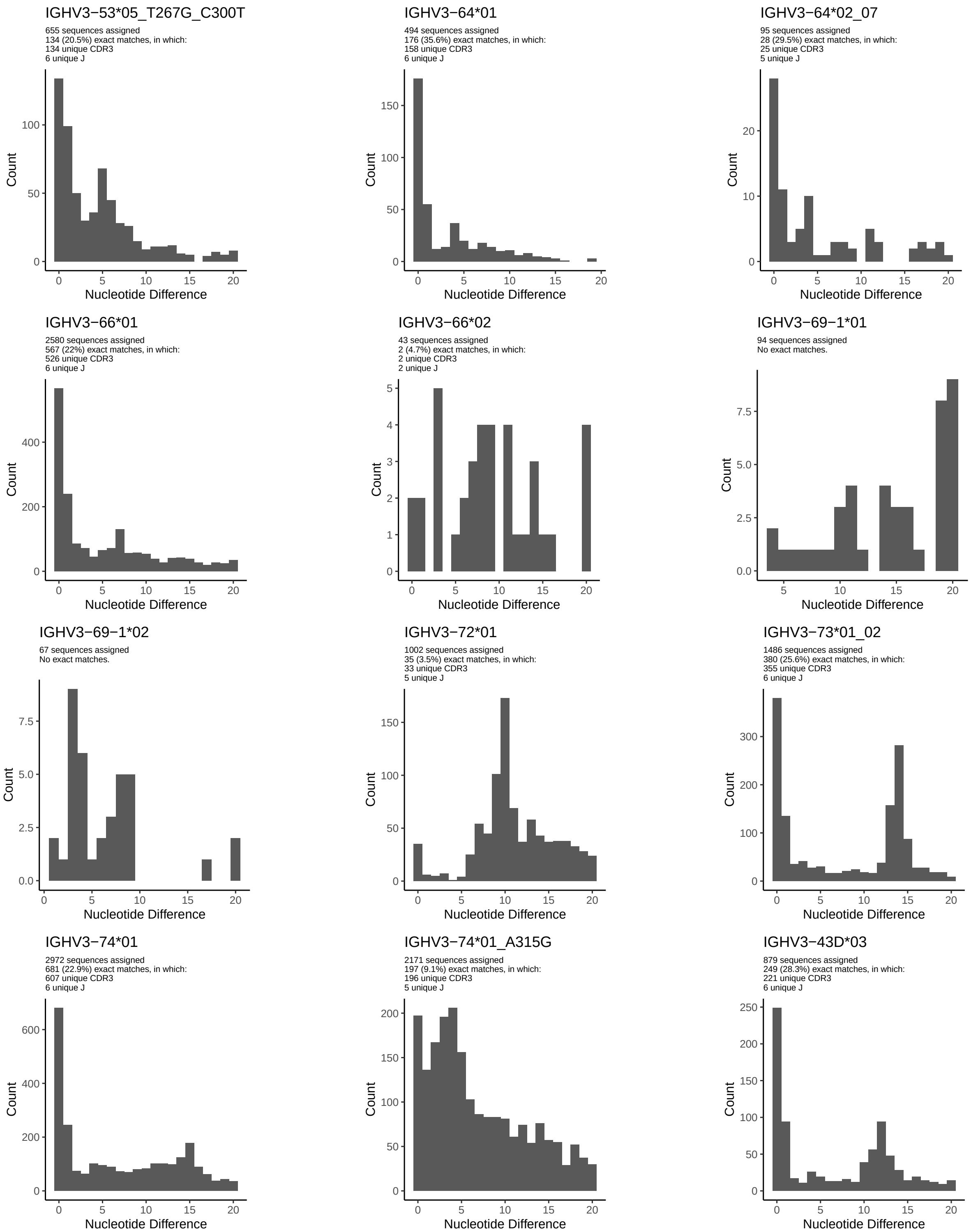


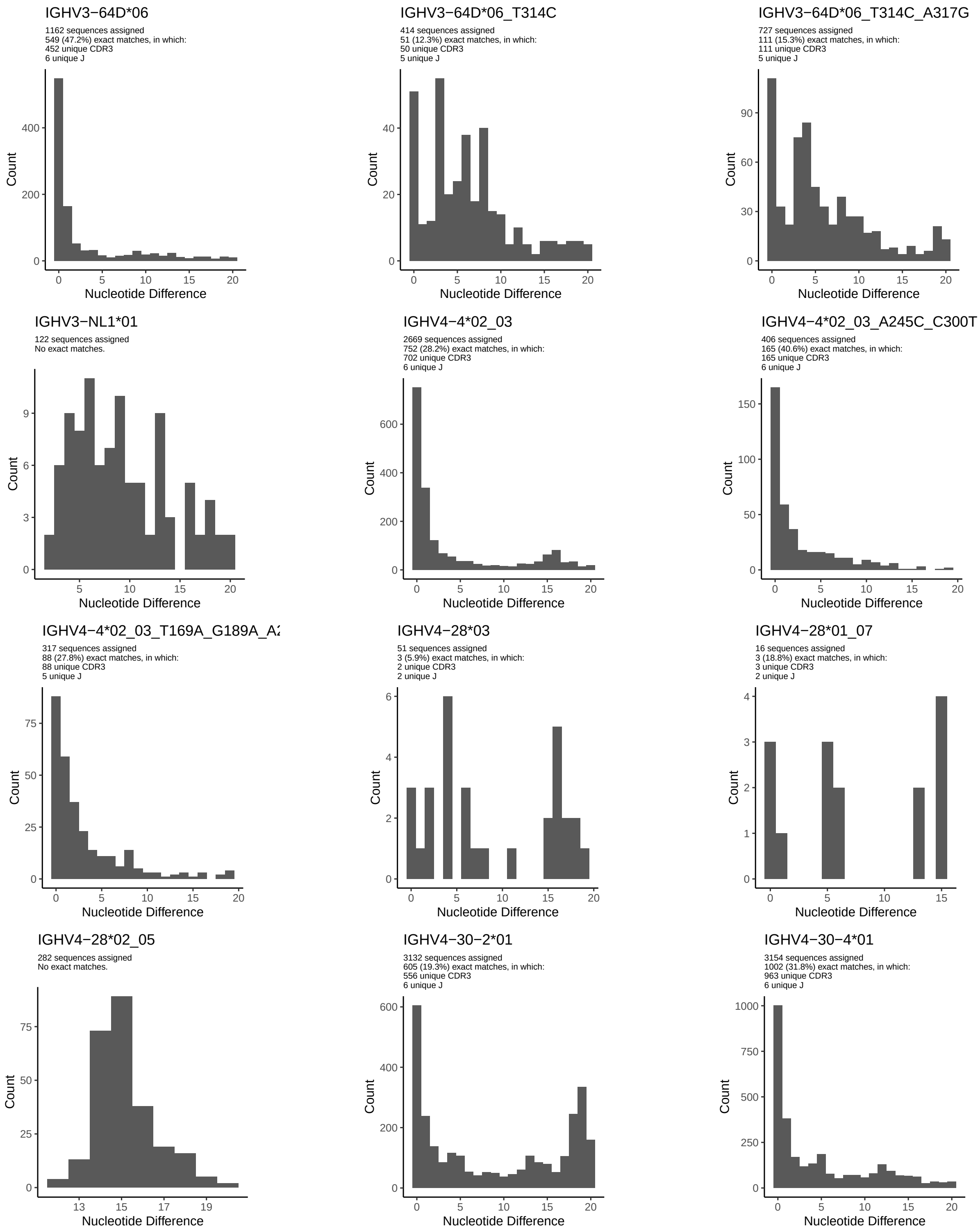

IGHV1−58*02
478 sequences assigned
165 (34.5%) exact matches, in which:
157 unique CDR3
6 unique J
IGHV1−58*01_03
605 sequences assigned
236 (39%) exact matches, in which:
225 unique CDR3
5 unique J
IGHV1−69−2*01
2 sequences assigned
No exact matches.
IGHV1−69*01_12_13
21563 sequences assigned
6898 (32%) exact matches, in which:
6527 unique CDR3
6 unique J
IGHV2−5*01
604 sequences assigned
177 (29.3%) exact matches, in which:
168 unique CDR3
6 unique J
IGHV2−5*02
8200 sequences assigned
151 (1.8%) exact matches, in which:
148 unique CDR3
6 unique J
IGHV2−26*01
547 sequences assigned
180 (32.9%) exact matches, in which:
174 unique CDR3
5 unique J
IGHV2−70*01
373 sequences assigned
129 (34.6%) exact matches, in which:
128 unique CDR3
5 unique J
IGHV2−70*15
815 sequences assigned
173 (21.2%) exact matches, in which:
170 unique CDR3
6 unique J
IGHV2−70D*04
1 sequences assigned
No exact matches.
IGHV3−7*01
8006 sequences assigned
1176 (14.7%) exact matches, in which:
1119 unique CDR3
6 unique J
IGHV3−7*03
12006 sequences assigned
1520 (12.7%) exact matches, in which:
1216 unique CDR3
6 unique J
Count
Nucleotide Difference

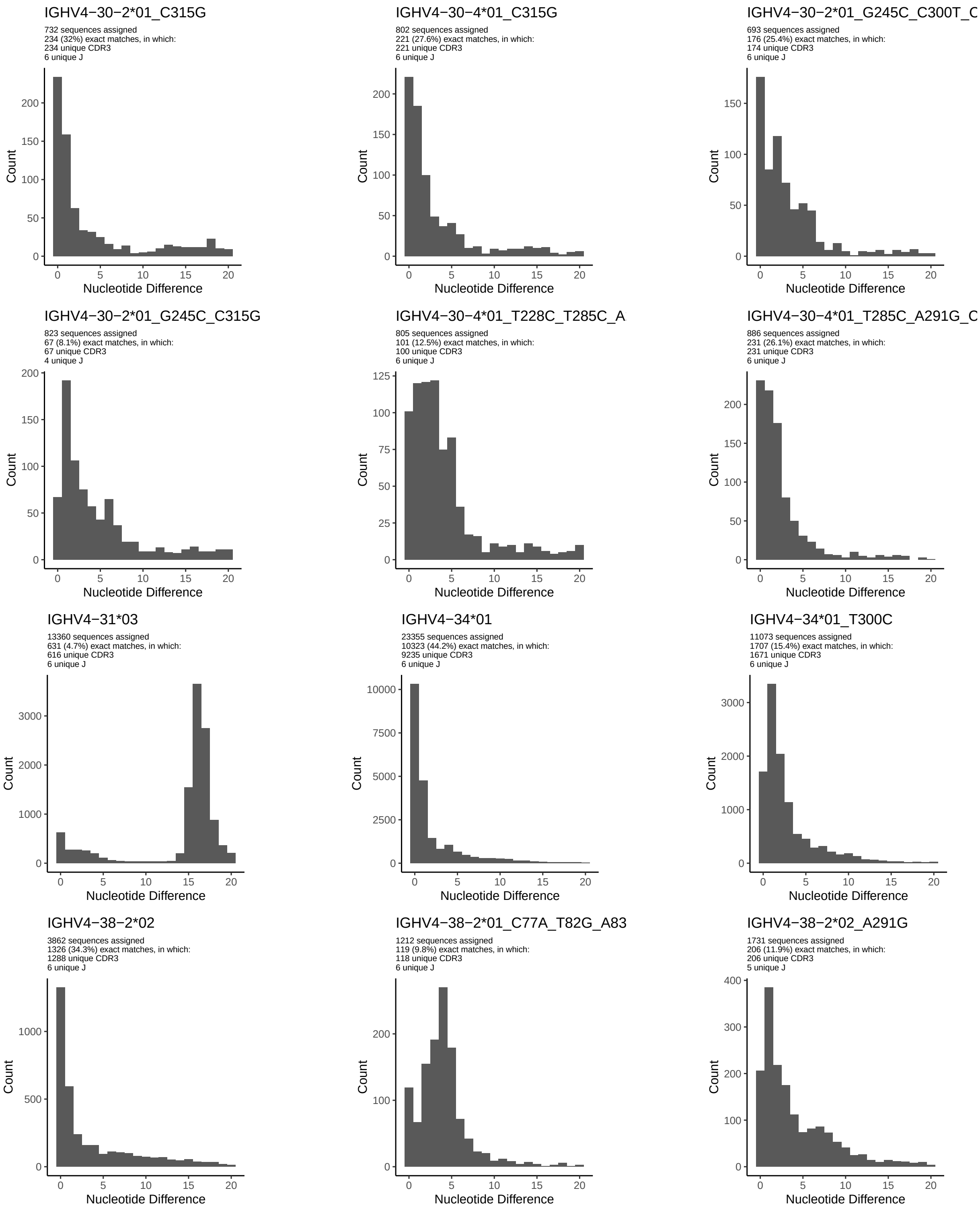

IGHV3−9*01
6884 sequences assigned
1031 (15%) exact matches, in which:
945 unique CDR3
6 unique J
IGHV3−11*01
4461 sequences assigned
801 (18%) exact matches, in which:
746 unique CDR3
6 unique J
IGHV3−11*04
1218 sequences assigned
117 (9.6%) exact matches, in which:
117 unique CDR3
4 unique J
IGHV3−11*06
4327 sequences assigned
709 (16.4%) exact matches, in which:
661 unique CDR3
6 unique J
IGHV3−13*01
815 sequences assigned
211 (25.9%) exact matches, in which:
193 unique CDR3
5 unique J
IGHV3−13*05
689 sequences assigned
88 (12.8%) exact matches, in which:
82 unique CDR3
5 unique J
IGHV3−13*06
116 sequences assigned
4 (3.4%) exact matches, in which:
4 unique CDR3
3 unique J
IGHV3−15*01
2641 sequences assigned
679 (25.7%) exact matches, in which:
628 unique CDR3
6 unique J
IGHV3−20*04
183 sequences assigned
1 (0.5%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−20*01_02
151 sequences assigned
55 (36.4%) exact matches, in which:
54 unique CDR3
4 unique J
IGHV3−21*01
10737 sequences assigned
2775 (25.8%) exact matches, in which:
2528 unique CDR3
6 unique J
IGHV3−23*01_04
37027 sequences assigned
4879 (13.2%) exact matches, in which:
4392 unique CDR3
6 unique J
Count
Nucleotide Difference

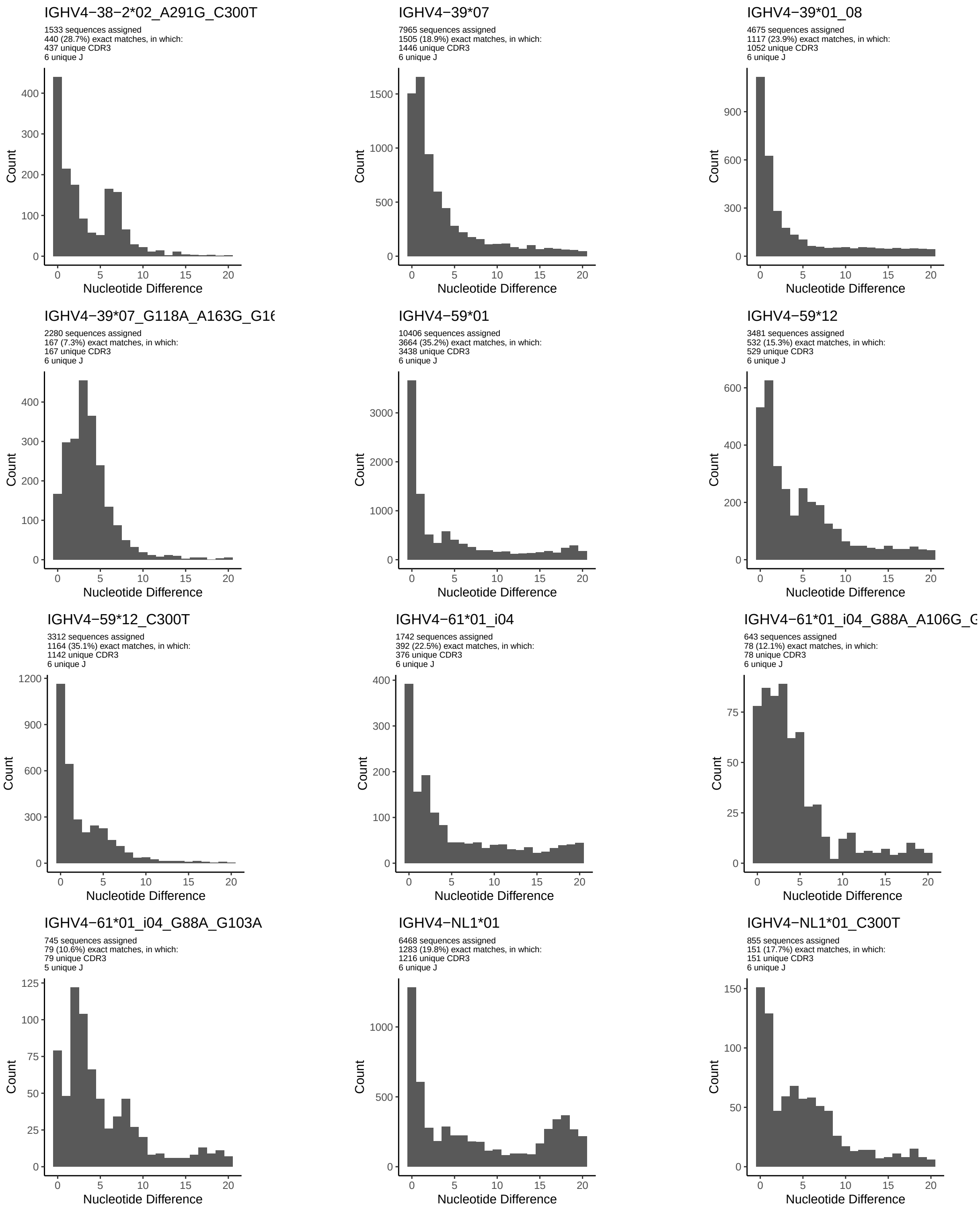

IGHV3−30−3*01
12794 sequences assigned
5257 (41.1%) exact matches, in which:
4799 unique CDR3
6 unique J
IGHV3−30*03
4585 sequences assigned
1399 (30.5%) exact matches, in which:
1366 unique CDR3
6 unique J
IGHV3−30*18
12141 sequences assigned
4059 (33.4%) exact matches, in which:
3724 unique CDR3
6 unique J
IGHV3−30−3*01_G317A
3131 sequences assigned
1028 (32.8%) exact matches, in which:
1009 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
3251 sequences assigned
453 (13.9%) exact matches, in which:
447 unique CDR3
6 unique J
IGHV3−30*03_T288C
3418 sequences assigned
528 (15.4%) exact matches, in which:
523 unique CDR3
6 unique J
IGHV3−30*20_G75C
2917 sequences assigned
748 (25.6%) exact matches, in which:
738 unique CDR3
6 unique J
IGHV3−33*01
2254 sequences assigned
930 (41.3%) exact matches, in which:
886 unique CDR3
6 unique J
IGHV3−33*08
1001 sequences assigned
255 (25.5%) exact matches, in which:
251 unique CDR3
6 unique J
IGHV3−33*01_C288T
773 sequences assigned
333 (43.1%) exact matches, in which:
333 unique CDR3
6 unique J
IGHV3−33*06_T300C_G303A
633 sequences assigned
102 (16.1%) exact matches, in which:
102 unique CDR3
5 unique J
IGHV3−35*01
2 sequences assigned
No exact matches.
Count
Nucleotide Difference

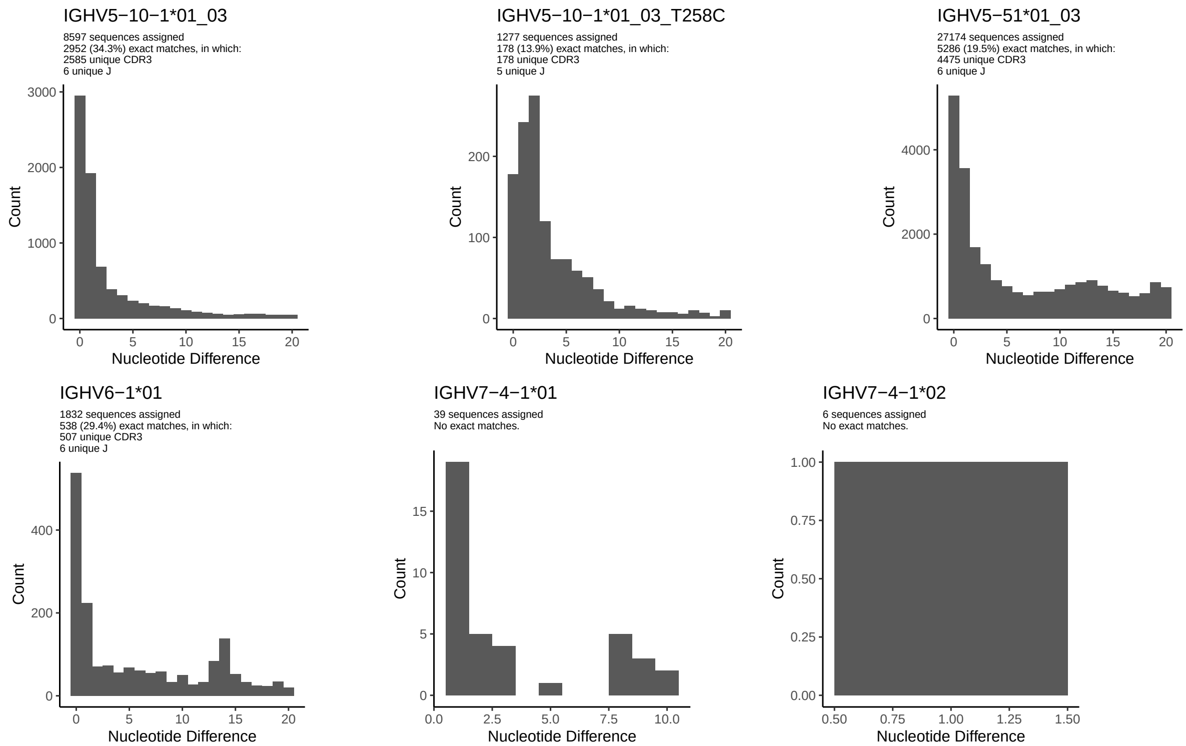

IGHV3−35*02
3 sequences assigned
2 (66.7%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−38−3*01
168 sequences assigned
31 (18.5%) exact matches, in which:
31 unique CDR3
3 unique J
IGHV3−38*02
3 sequences assigned
No exact matches.
IGHV3−38*03
9 sequences assigned
No exact matches.
IGHV3−43*01
570 sequences assigned
155 (27.2%) exact matches, in which:
147 unique CDR3
5 unique J
IGHV3−47*02
2 sequences assigned
No exact matches.
IGHV3−48*02
2701 sequences assigned
735 (27.2%) exact matches, in which:
683 unique CDR3
6 unique J
IGHV3−48*04
5794 sequences assigned
1119 (19.3%) exact matches, in which:
1032 unique CDR3
6 unique J
IGHV3−48*01_G247T_T256A_A25
1379 sequences assigned
123 (8.9%) exact matches, in which:
123 unique CDR3
6 unique J
IGHV3−49*03_05
983 sequences assigned
386 (39.3%) exact matches, in which:
363 unique CDR3
6 unique J
IGHV3−53*04
689 sequences assigned
255 (37%) exact matches, in which:
235 unique CDR3
6 unique J
IGHV3−53*01_02
2755 sequences assigned
554 (20.1%) exact matches, in which:
514 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−53*05_T267G_C300T
655 sequences assigned
134 (20.5%) exact matches, in which:
134 unique CDR3
6 unique J
IGHV3−64*01
494 sequences assigned
176 (35.6%) exact matches, in which:
158 unique CDR3
6 unique J
IGHV3−64*02_07
95 sequences assigned
28 (29.5%) exact matches, in which:
25 unique CDR3
5 unique J
IGHV3−66*01
2580 sequences assigned
567 (22%) exact matches, in which:
526 unique CDR3
6 unique J
IGHV3−66*02
43 sequences assigned
2 (4.7%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−69−1*01
94 sequences assigned
No exact matches.
IGHV3−69−1*02
67 sequences assigned
No exact matches.
IGHV3−72*01
1002 sequences assigned
35 (3.5%) exact matches, in which:
33 unique CDR3
5 unique J
IGHV3−73*01_02
1486 sequences assigned
380 (25.6%) exact matches, in which:
355 unique CDR3
6 unique J
IGHV3−74*01
2972 sequences assigned
681 (22.9%) exact matches, in which:
607 unique CDR3
6 unique J
IGHV3−74*01_A315G
2171 sequences assigned
197 (9.1%) exact matches, in which:
196 unique CDR3
5 unique J
IGHV3−43D*03
879 sequences assigned
249 (28.3%) exact matches, in which:
221 unique CDR3
6 unique J
Count
Nucleotide Difference

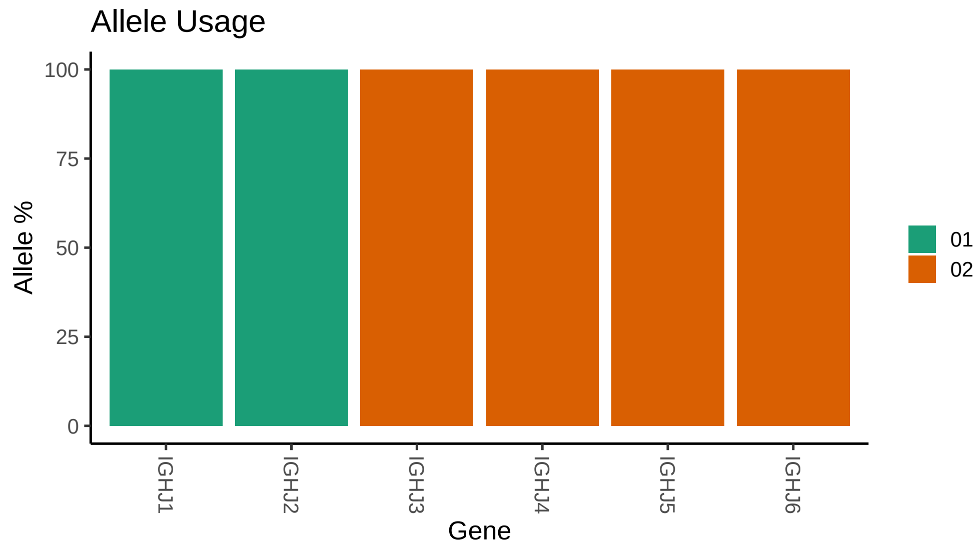

IGHV3−64D*06
1162 sequences assigned
549 (47.2%) exact matches, in which:
452 unique CDR3
6 unique J
IGHV3−64D*06_T314C
414 sequences assigned
51 (12.3%) exact matches, in which:
50 unique CDR3
5 unique J
IGHV3−64D*06_T314C_A317G
727 sequences assigned
111 (15.3%) exact matches, in which:
111 unique CDR3
5 unique J
IGHV3−NL1*01
122 sequences assigned
No exact matches.
IGHV4−4*02_03
2669 sequences assigned
752 (28.2%) exact matches, in which:
702 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
406 sequences assigned
165 (40.6%) exact matches, in which:
165 unique CDR3
6 unique J
IGHV4−4*02_03_T169A_G189A_A2
317 sequences assigned
88 (27.8%) exact matches, in which:
88 unique CDR3
5 unique J
IGHV4−28*03
51 sequences assigned
3 (5.9%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−28*01_07
16 sequences assigned
3 (18.8%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV4−28*02_05
282 sequences assigned
No exact matches.
IGHV4−30−2*01
3132 sequences assigned
605 (19.3%) exact matches, in which:
556 unique CDR3
6 unique J
IGHV4−30−4*01
3154 sequences assigned
1002 (31.8%) exact matches, in which:
963 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−30−2*01_C315G
732 sequences assigned
234 (32%) exact matches, in which:
234 unique CDR3
6 unique J
IGHV4−30−4*01_C315G
802 sequences assigned
221 (27.6%) exact matches, in which:
221 unique CDR3
6 unique J
IGHV4−30−2*01_G245C_C300T_C
693 sequences assigned
176 (25.4%) exact matches, in which:
174 unique CDR3
6 unique J
IGHV4−30−2*01_G245C_C315G
823 sequences assigned
67 (8.1%) exact matches, in which:
67 unique CDR3
4 unique J
IGHV4−30−4*01_T228C_T285C_A
805 sequences assigned
101 (12.5%) exact matches, in which:
100 unique CDR3
6 unique J
IGHV4−30−4*01_T285C_A291G_C
886 sequences assigned
231 (26.1%) exact matches, in which:
231 unique CDR3
6 unique J
IGHV4−31*03
13360 sequences assigned
631 (4.7%) exact matches, in which:
616 unique CDR3
6 unique J
IGHV4−34*01
23355 sequences assigned
10323 (44.2%) exact matches, in which:
9235 unique CDR3
6 unique J
IGHV4−34*01_T300C
11073 sequences assigned
1707 (15.4%) exact matches, in which:
1671 unique CDR3
6 unique J
IGHV4−38−2*02
3862 sequences assigned
1326 (34.3%) exact matches, in which:
1288 unique CDR3
6 unique J
IGHV4−38−2*01_C77A_T82G_A83
1212 sequences assigned
119 (9.8%) exact matches, in which:
118 unique CDR3
6 unique J
IGHV4−38−2*02_A291G
1731 sequences assigned
206 (11.9%) exact matches, in which:
206 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV4−38−2*02_A291G_C300T
1533 sequences assigned
440 (28.7%) exact matches, in which:
437 unique CDR3
6 unique J
IGHV4−39*07
7965 sequences assigned
1505 (18.9%) exact matches, in which:
1446 unique CDR3
6 unique J
IGHV4−39*01_08
4675 sequences assigned
1117 (23.9%) exact matches, in which:
1052 unique CDR3
6 unique J
IGHV4−39*07_G118A_A163G_G1
2280 sequences assigned
167 (7.3%) exact matches, in which:
167 unique CDR3
6 unique J
IGHV4−59*01
10406 sequences assigned
3664 (35.2%) exact matches, in which:
3438 unique CDR3
6 unique J
IGHV4−59*12
3481 sequences assigned
532 (15.3%) exact matches, in which:
529 unique CDR3
6 unique J
IGHV4−59*12_C300T
3312 sequences assigned
1164 (35.1%) exact matches, in which:
1142 unique CDR3
6 unique J
IGHV4−61*01_i04
1742 sequences assigned
392 (22.5%) exact matches, in which:
376 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_A106G_
643 sequences assigned
78 (12.1%) exact matches, in which:
78 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A
745 sequences assigned
79 (10.6%) exact matches, in which:
79 unique CDR3
5 unique J
IGHV4−NL1*01
6468 sequences assigned
1283 (19.8%) exact matches, in which:
1216 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
855 sequences assigned
151 (17.7%) exact matches, in which:
151 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV5−10−1*01_03
8597 sequences assigned
2952 (34.3%) exact matches, in which:
2585 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV5−10−1*01_03_T258C
1277 sequences assigned
178 (13.9%) exact matches, in which:
178 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV5−51*01_03
27174 sequences assigned
5286 (19.5%) exact matches, in which:
4475 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV6−1*01
1832 sequences assigned
538 (29.4%) exact matches, in which:
507 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV7−4−1*01
39 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV7−4−1*02
6 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 002/M64 002/M64 002/M64 002/70/33f0d28c66d6d84a9
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 002/M64 002/M64 002/M64 002/70/33f0d28c6
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 002/M64 002/M64 002/M64 002/70/33f0d28c66d6d84
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 18 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 18 04_T81A_T90C_A106G_G112T_G113A_C117G_A118C_G119A: replacing
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 20_G75C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_G317A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 01_C288T: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 01_G247T_T256A_A258G_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 66 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 05_T267G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 64D 06: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T314C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T314C_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_T169A_G189A_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C300T_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_G245C_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T285C_A291G_C300T_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_T228C_T285C_A291G_C300T_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_C77A_T82G_A83G_C84G_A88T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 07_G118A_A163G_G164A_T165A_T169A_T172C_G189A_T196A_C300T: re
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_A106G_G107A_C120T_G129C_A147C_T165C: replacing w
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV5 10 1 01_03: replacing with gap
```

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##
##
## Unexpected N in novel sequence IGHV5 10 1 01_03_T258C: replacing with gap
```